

5

Balance Laws

In this chapter we state various axioms which form the basis for a thermo-mechanical theory of continuum bodies. These axioms provide a set of balance laws which describe how the mass, momentum, energy and entropy of a body change in time under prescribed external influences. We first state these laws in global or integral form, then derive various corresponding local statements, primarily in the form of partial differential equations. The balance laws stated here apply to all bodies regardless of their constitution. In Chapters 6–9 these laws are specialized to various classes of bodies with specific material properties, via constitutive models.

The important ideas in this chapter are: (i) the balance laws of mass, momentum, energy and entropy for continuum bodies; (ii) the difference between the integral form of a law and its local Eulerian and Lagrangian forms; (iii) the axiom of material frame-indifference and its role in constitutive modeling; (iv) the idea of a material constraint and its implications for the stress field in a body; (v) the balance laws relevant to the isothermal modeling of continuum bodies.

5.1 Motivation

In order to motivate the contents of this chapter it is useful to recall some basic ideas from the mechanics of particle systems. To this end, we consider a system of N particles with masses m_i and positions $\mathbf{x}_i$ as illustrated in Figure 5.1. It will be helpful to think of these particles as the atoms making up a continuum body. We suppose that every distinct pair of particles ($j \neq i$) interacts with an energy U_{ij} , which is assumed to be symmetric in the sense that $U_{ij} = U_{ji}$, and we denote the interaction force of particle j on i by $\mathbf{f}_{ij}^{\text{int}} = -\nabla^{\mathbf{x}_i} U_{ij}$. We further suppose that

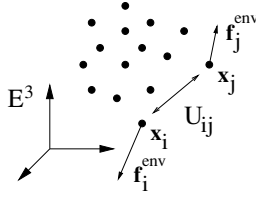


Fig. 5.1 Illustration of a particle system. $\mathbf{x}_i$ denotes the position of particle i , $\mathbf{f}_i^{\text{env}}$ denotes an environmental force on particle i , and U_{ij} denotes a pairwise interaction energy between particles i and j .

each particle is subject to an environmental force $\mathbf{f}_i^{\text{env}}$. For simplicity, we assume that no other forces act on or within the system.

A complete description of the motion of the particle system is given by the equations (no summation convention)

$$\dot{m}_i = 0, \quad m_i \ddot{\mathbf{x}}_i = \mathbf{f}_i^{\text{env}} + \sum_{\substack{j=1 \\ j \neq i}}^N \mathbf{f}_{ij}^{\text{int}}, \quad \forall i = 1, \dots, N. \quad (5.1)$$

The first equation states that the mass of each particle is constant. The second equation states that each particle obeys **Newton's Second Law**. That is, the mass of each particle times its acceleration is equal to the resultant force on it.

Under suitable assumptions on the interaction energies U_{ij} , the above equations can be used to derive balance laws for mass, linear momentum, angular momentum and energy for any subset of the particle system. To this end, consider an arbitrary subset I of $\{1, \dots, N\}$ and let Ω_t denote the current configuration of the particles with labels $i \in I$. To the configuration Ω_t at any time $t \geq 0$ we associate a total mass $M[\Omega_t]$, linear momentum $\mathbf{l}[\Omega_t]$, angular momentum $\mathbf{j}[\Omega_t]$, internal energy $U[\Omega_t]$ and kinetic energy $K[\Omega_t]$ as follows

$$\begin{aligned} M[\Omega_t] &= \sum_{i \in I} m_i, & \mathbf{l}[\Omega_t] &= \sum_{i \in I} m_i \dot{\mathbf{x}}_i, & \mathbf{j}[\Omega_t] &= \sum_{i \in I} \mathbf{x}_i \times m_i \dot{\mathbf{x}}_i, \\ U[\Omega_t] &= \sum_{\substack{i, j \in I \\ i < j}} U_{ij}, & K[\Omega_t] &= \sum_{i \in I} \frac{1}{2} m_i |\dot{\mathbf{x}}_i|^2. \end{aligned} \quad (5.2)$$

Assuming U_{ij} depends on $\mathbf{x}_i$ and $\mathbf{x}_j$ only through the pairwise distance $|\mathbf{x}_i - \mathbf{x}_j|$ and interactions are symmetric as described before, we deduce the following equations from (5.1)

$$\frac{d}{dt} M[\Omega_t] = 0, \quad (5.3)$$

$$\frac{d}{dt}I[\Omega_t] = \sum_{i \in I} \left[\mathbf{f}_i^{\text{env}} + \sum_{j \notin I} \mathbf{f}_{ij}^{\text{int}} \right], \quad (5.4)$$

$$\frac{d}{dt}\mathbf{j}[\Omega_t] = \sum_{i \in I} \mathbf{x}_i \times \left[\mathbf{f}_i^{\text{env}} + \sum_{j \notin I} \mathbf{f}_{ij}^{\text{int}} \right], \quad (5.5)$$

$$\frac{d}{dt}(U[\Omega_t] + K[\Omega_t]) = \sum_{i \in I} \dot{\mathbf{x}}_i \cdot \left[\mathbf{f}_i^{\text{env}} + \sum_{j \notin I} \mathbf{f}_{ij}^{\text{int}} \right]. \quad (5.6)$$

These equations state that the mass of Ω_t does not change with time, the rate of change of linear momentum is equal to the resultant external force on Ω_t , the rate of change of angular momentum is equal to the resultant external torque on Ω_t and the rate of change of the sum of internal and kinetic energy is equal to the power of external forces on Ω_t (see Exercises 1 and 2).

Equations (5.3)–(5.6) are typically referred to as **balance laws**. They describe how the mass, linear momentum, angular momentum and energy of Ω_t may be conserved, increased or decreased depending on external influences. For example, consider the case when Ω_t corresponds to the entire system of particles, equivalently $I = \{1, \dots, N\}$, and suppose all environmental forces are zero. In this case, equations (5.3)–(5.6) imply that the mass, linear momentum, angular momentum and energy of Ω_t would all be conserved throughout any motion. That is, these quantities would remain constant.

The four balance laws (5.3)–(5.6) provide a fundamental starting point in the modeling of continuum bodies. The balance laws for mass, linear momentum and angular momentum can be generalized to continuum bodies in a straightforward manner: we essentially replace the summation by an integral. The balance law for energy can also be generalized; however, the situation is more complicated. Continuum modeling cannot represent the detail that occurs at length scales comparable to, or shorter than, inter-atomic spacings. In particular, the velocity at a point in a continuum body is to be interpreted as the average velocity of individual atoms in the vicinity of that point.

Fluctuations in the velocities of the individual atoms about the mean are not represented in the continuum velocity field. They are associated with time and length scales on which the model no longer applies. To account for situations in which the energy of these fluctuations is significant, it is necessary to introduce new concepts at the continuum

level: **temperature**, to measure the size of the velocity fluctuations, and **heat**, to measure the energy of these fluctuations. Temperature and heat will play an important role in the generalization of (5.6) to continuum bodies.

When suitably generalized, the four balance laws (5.3)–(5.6) form the basis of a thermo-mechanical theory of continuum bodies. In this case, the four laws are essential postulates. That is, while (5.3)–(5.6) are all implied by (5.1) for systems of particles under special assumptions, there is no such implication in general when the assumptions are removed. For this reason, we simply postulate these laws in the continuum case and note that their validity has been verified by experiment. A fifth, purely thermal, balance law will also be postulated in the continuum case. This law, which is motivated by the thermodynamics of homogeneous processes, involves the concept of entropy and will take the form of an inequality. The four balance laws corresponding to (5.3)–(5.6), together with the entropy inequality, will provide a complete thermo-mechanical theory of continuum bodies.

5.2 Balance Laws in Integral Form

In this section we state the balance laws for mass, momentum, energy and entropy as they apply to any open subset of a continuum body. We begin with a statement of the conservation of mass, and then proceed to the laws of inertia which connect the rates of change of the total linear and angular momentum of a body to external forces applied to it. Introducing the notion of internal energy and heating of a continuum body, and the net working of external forces, we state the laws of thermodynamics which describe how various forms of energy, such as thermal and mechanical, are interconverted.

5.2.1 Conservation of Mass and Laws of Inertia

Consider a continuum body with reference configuration B undergoing a motion $\varphi : B \times [0, \infty) \rightarrow \mathbb{E}^3$. Let B_t denote the current configuration, and let $\rho(\mathbf{x}, t)$ denote the mass density and $\mathbf{v}(\mathbf{x}, t)$ the velocity at a point $\mathbf{x} \in B_t$ at time $t \geq 0$. Moreover, let Ω_t denote an arbitrary open subset of B_t . Motivated by (5.2) we define the **mass** of Ω_t by

$$\text{mass}[\Omega_t] = \int_{\Omega_t} \rho(\mathbf{x}, t) \, dV_{\mathbf{x}},$$

the **linear momentum** by

$$\mathbf{l}[\Omega_t] = \int_{\Omega_t} \rho(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t) dV_{\mathbf{x}},$$

and the **angular momentum** with respect to a point $\mathbf{z}$ by

$$\mathbf{j}[\Omega_t]_{\mathbf{z}} = \int_{\Omega_t} (\mathbf{x} - \mathbf{z}) \times \rho(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t) dV_{\mathbf{x}}.$$

In the absence of chemical reactions and relativistic effects, the balance law stated in (5.3) is generalized to continuum bodies as follows.

Axiom 5.1 Conservation of Mass. *The mass of any open subset of a continuum body does not change as the body changes place and shape. That is*

$$\frac{d}{dt} \text{mass}[\Omega_t] = 0, \quad \forall \Omega_t \subseteq B_t.$$

□

Suppose Ω_t is subject to an external traction field $\mathbf{t}(\mathbf{x}, t)$ and an external body force field per unit mass $\mathbf{b}(\mathbf{x}, t)$ (see Chapter 3). Then the resultant force on Ω_t is

$$\mathbf{r}[\Omega_t] = \int_{\Omega_t} \rho(\mathbf{x}, t) \mathbf{b}(\mathbf{x}, t) dV_{\mathbf{x}} + \int_{\partial\Omega_t} \mathbf{t}(\mathbf{x}, t) dA_{\mathbf{x}},$$

and the resultant torque on Ω_t with respect to $\mathbf{z}$ is

$$\boldsymbol{\tau}[\Omega_t]_{\mathbf{z}} = \int_{\Omega_t} (\mathbf{x} - \mathbf{z}) \times \rho(\mathbf{x}, t) \mathbf{b}(\mathbf{x}, t) dV_{\mathbf{x}} + \int_{\partial\Omega_t} (\mathbf{x} - \mathbf{z}) \times \mathbf{t}(\mathbf{x}, t) dA_{\mathbf{x}}.$$

The balance laws stated in (5.4) and (5.5) are generalized to continuum bodies as follows.

Axiom 5.2 Laws of Inertia. *With respect to a fixed frame of reference, the rate of change of linear momentum of any open subset of a continuum body equals the resultant force applied to it, and the rate of change of angular momentum about the origin equals the resultant torque about the origin. That is*

$$\frac{d}{dt} \mathbf{l}[\Omega_t] = \mathbf{r}[\Omega_t] \quad \text{and} \quad \frac{d}{dt} \mathbf{j}[\Omega_t]_{\mathbf{o}} = \boldsymbol{\tau}[\Omega_t]_{\mathbf{o}}, \quad \forall \Omega_t \subseteq B_t. \quad (5.7)$$

□

The above laws are postulated to hold for every motion and every subset of a continuum body. Beginning from these laws it is also possible to derive alternative forms of the angular momentum equation. In particular, from (5.7) we deduce that the angular momentum of Ω_t also satisfies the equations

$$\frac{d}{dt} \mathbf{j}[\Omega_t]_{\mathbf{z}} = \boldsymbol{\tau}[\Omega_t]_{\mathbf{z}} \quad \text{and} \quad \frac{d}{dt} \mathbf{j}[\Omega_t]_{\mathbf{x}_{\text{com}}} = \boldsymbol{\tau}[\Omega_t]_{\mathbf{x}_{\text{com}}}, \quad (5.8)$$

where $\mathbf{z}$ is any fixed point and $\mathbf{x}_{\text{com}}$ is the generally time-dependent mass center of Ω_t (see Exercises 3 and 5). This is analogous to the equilibrium situation outlined in Chapter 3. The forms of the angular momentum equation in (5.8) are typically used in the study of rigid bodies. The linear momentum equation in (5.7) can also be phrased in terms of the mass center of Ω_t (see Exercise 4).

5.2.2 First and Second Laws of Thermodynamics

5.2.2.1 Temperature and Heat

In the most basic sense, **temperature** is a physical property of matter that is based on our natural perception of hot and cold. In the modeling of continuum bodies we assume that a temperature is defined at each point in a body at each time. In particular, we assume the existence of an absolute **temperature field** $\theta(\mathbf{x}, t) > 0$ defined for $\mathbf{x} \in B_t$ and $t \geq 0$. The temperature at a point in a continuum body is to be interpreted as a measure of the velocity fluctuations of individual atoms in the vicinity of that point. Thus, while continuum velocity is a measure of the mean, continuum temperature is a measure of the variation of atomistic velocities near a point.

By the **thermal energy** or **heat content** of a body we mean an energy associated with the velocity fluctuations of individual atoms in the body. In particular, heat is an energy associated with temperature. Physical experience tells us that material bodies are capable of interconverting heat and mechanical work. Deforming a body can affect its temperature and thereby its heat content. Similarly, adding heat to a body by raising its temperature may set the body in motion, for example it may expand. Because of this interconversion, heat may be expressed in the same units as mechanical work.

The heat content of a body can be affected in two basic ways. First, heat can be produced or consumed throughout the volume of a body by mechanical, chemical or electromagnetic mechanisms. For example, an

electrical current can produce heat as it flows through the volume of an isolated body. Second, heat can be transferred from one body to another through physical contact or through thermal radiation at a surface. For example, when a warm body is put in physical contact with a cool one, experience tells us that the warm body will get cooler (lose heat) and the cool body will get warmer (gain heat). In this case there is a transfer of heat per unit area of the contact surface. Transfer through contact can be further classified as conduction or convection depending on the circumstances and the nature of the bodies involved.

Given an arbitrary open subset Ω_t of B_t we define the **net heating** $Q[\Omega_t]$ as the rate per unit time at which heat is being added to Ω_t . In analogy with our treatment of forces (see Chapter 3), we assume that the net heating can be decomposed into a body heating $Q_b[\Omega_t]$ and a surface heating $Q_s[\Omega_t]$ such that

$$Q[\Omega_t] = Q_b[\Omega_t] + Q_s[\Omega_t]. \quad (5.9)$$

We assume the body heating $Q_b[\Omega_t]$ is a continuous function of volume. In particular, we assume there exists a **heat supply field** per unit volume $\hat{r}(\mathbf{x}, t) \in \mathbb{R}$ such that

$$Q_b[\Omega_t] = \int_{\Omega_t} \hat{r}(\mathbf{x}, t) dV_{\mathbf{x}}.$$

For convenience, we define a heat supply field per unit mass $r(\mathbf{x}, t) \in \mathbb{R}$ by

$$r(\mathbf{x}, t) = \rho(\mathbf{x}, t)^{-1} \hat{r}(\mathbf{x}, t),$$

so that

$$Q_b[\Omega_t] = \int_{\Omega_t} \rho(\mathbf{x}, t) r(\mathbf{x}, t) dV_{\mathbf{x}}. \quad (5.10)$$

We assume the surface heating $Q_s[\Omega_t]$ is a continuous function of surface area. In particular, we assume there exists a **heat transfer field** per unit area $h(\mathbf{x}, t) \in \mathbb{R}$ such that

$$Q_s[\Omega_t] = \int_{\partial\Omega_t} h(\mathbf{x}, t) dA_{\mathbf{x}}. \quad (5.11)$$

Moreover, we assume the heat transfer field is of the form

$$h(\mathbf{x}, t) = -\mathbf{q}(\mathbf{x}, t) \cdot \mathbf{n}(\mathbf{x}), \quad (5.12)$$

where $\mathbf{n}$ is the outward unit normal field for the surface $\partial\Omega_t$ and $\mathbf{q}$ is the Fourier–Stokes **heat flux vector field** in B_t .

The direction and intensity of heat flow across any surface in a body is determined by the vector field $\mathbf{q}$. The negative sign in (5.12) is due to the fact that $\mathbf{n}$ is chosen to be the outward as opposed to inward unit normal. In particular, from (5.12) we deduce that h is positive when $\mathbf{q}$ points into Ω_t . The relation between the surface heat transfer and the heat flux vector is analogous to the relation between the surface traction and the Cauchy stress tensor (see Result 3.3). In particular, it can be shown that, under mild assumptions, both traction and heat transfer must necessarily be linear in the surface normal.

Substituting (5.10), (5.11) and (5.12) into (5.9) we obtain

$$Q[\Omega_t] = \int_{\Omega_t} \rho(\mathbf{x}, t) r(\mathbf{x}, t) dV_{\mathbf{x}} - \int_{\partial\Omega_t} \mathbf{q}(\mathbf{x}, t) \cdot \mathbf{n}(\mathbf{x}) dA_{\mathbf{x}}. \quad (5.13)$$

Notice that heat is absorbed or released from Ω_t depending on whether $Q[\Omega_t]$ is positive or negative.

5.2.2.2 Kinetic Energy, Power of External Forces and Net Working

Consider an arbitrary open subset Ω_t of B_t with external traction field $\mathbf{t}(\mathbf{x}, t)$ and external body force field per unit mass $\mathbf{b}(\mathbf{x}, t)$. We define the **kinetic energy** of Ω_t by

$$K[\Omega_t] = \int_{\Omega_t} \frac{1}{2} \rho(\mathbf{x}, t) |\mathbf{v}(\mathbf{x}, t)|^2 dV_{\mathbf{x}},$$

and the **power of external forces** on Ω_t by

$$\mathcal{P}[\Omega_t] = \int_{\Omega_t} \rho(\mathbf{x}, t) \mathbf{b}(\mathbf{x}, t) \cdot \mathbf{v}(\mathbf{x}, t) dV_{\mathbf{x}} + \int_{\partial\Omega_t} \mathbf{t}(\mathbf{x}, t) \cdot \mathbf{v}(\mathbf{x}, t) dA_{\mathbf{x}}.$$

By the **net working** $\mathcal{W}[\Omega_t]$ of external forces on Ω_t , we mean the mechanical power delivered by these forces that is not used up in producing motion, that is

$$\mathcal{W}[\Omega_t] = \mathcal{P}[\Omega_t] - \frac{d}{dt} K[\Omega_t]. \quad (5.14)$$

If $\mathcal{W}[\Omega_t] = 0$ throughout a time interval, then all the mechanical energy delivered to the body by external forces is used to produce motion. In particular, it is converted to kinetic energy. If $\mathcal{W}[\Omega_t] > 0$ throughout a time interval, then some of the mechanical energy delivered to the body is stored or converted to some form other than kinetic energy. Conversely, if $\mathcal{W}[\Omega_t] < 0$, then energy stored in the body is released as work against the external forces or as motion.

5.2.2.3 Internal Energy and The First Law

The energy content of a continuum body not associated with kinetic energy is called **internal energy**. The internal energy of a body represents a store of energy which may be increased in various ways, for example by adding heat to the body or performing mechanical work on it. Once stored, internal energy can be released in any form, for example heat, mechanical work or motion.

For our purposes, we assume the internal energy of a body consists only of thermal (heat) and mechanical (elastic) energy. In particular, we neglect such things as chemical and electromagnetic energy, which in general would also contribute to the internal energy. Neglecting these other contributions is tantamount to assuming that they are constants independent of any net heating and working. In the analysis of some problems it is often possible and convenient to introduce a potential energy associated with external forces. Such an energy should not be confused with internal energy. A brief discussion of potential energy is given below.

Given an arbitrary open subset Ω_t of B_t we denote its internal energy by $U[\Omega_t]$. We assume $U[\Omega_t]$ is a continuous function of volume. In particular, we assume there exists an **internal energy density field** per unit volume $\hat{\phi}(\mathbf{x}, t) \in \mathbf{R}$ such that

$$U[\Omega_t] = \int_{\Omega_t} \hat{\phi}(\mathbf{x}, t) dV_{\mathbf{x}}.$$

For convenience, we define an internal energy density field per unit mass $\phi(\mathbf{x}, t) \in \mathbf{R}$ by

$$\phi(\mathbf{x}, t) = \rho^{-1}(\mathbf{x}, t) \hat{\phi}(\mathbf{x}, t),$$

so that

$$U[\Omega_t] = \int_{\Omega_t} \rho(\mathbf{x}, t) \phi(\mathbf{x}, t) dV_{\mathbf{x}}.$$

Considering only thermo-mechanical energy, the balance law stated in (5.6) is generalized to continuum bodies as follows.

Axiom 5.3 First Law of Thermodynamics. *The rate of change of internal energy of any open subset of a continuum body equals the sum of the net heating and net working applied to it. That is*

$$\frac{d}{dt} U[\Omega_t] = Q[\Omega_t] + \mathcal{W}[\Omega_t], \quad \forall \Omega_t \subseteq B_t.$$

In view of (5.14), we have

$$\frac{d}{dt}(U[\Omega_t] + K[\Omega_t]) = Q[\Omega_t] + \mathcal{P}[\Omega_t], \quad \forall \Omega_t \subseteq B_t. \quad (5.15)$$

□

There are various fundamental differences between the continuum energy balance law (5.15) and the corresponding particle law (5.6). For example, the kinetic energy in a continuum model is an energy associated with a local mean velocity of individual particles (atoms). In contrast, the kinetic energy in a particle model is an energy associated with individual velocities. Furthermore, the internal energy in a continuum model has both mechanical (elastic) and thermal (heat) contributions. In contrast, the internal energy in a particle model has only a mechanical contribution. In general, the energy balance law (5.6) for particles does not explicitly contain any quantities related to heat. Similar fundamental differences can also be identified between the continuum and particle balance laws pertaining to momentum.

In some cases, the power of external forces can be written in the form

$$\mathcal{P}[\Omega_t] = -\frac{d}{dt}G[\Omega_t],$$

where $G[\Omega_t]$ is called a **potential energy for external forces**. In this special case the energy balance law (5.15) becomes

$$\frac{d}{dt}(U[\Omega_t] + K[\Omega_t] + G[\Omega_t]) = Q[\Omega_t].$$

This says that the rate of change of the total energy $U + K + G$ is equal to the net heating Q . When the net heating is zero, this total energy is conserved throughout a motion.

5.2.2.4 Entropy and the Second Law

The energy balance law provides a relation between the rate of change of internal energy and the net heating and working of external influences on a body. While this relation implies that the rates must always be balanced, it places no fundamental restriction on the rates themselves. For example, in any time interval in which the net working is zero, the energy balance law does not limit the rate at which a body may absorb heat and store it as internal energy. Similarly, it does not limit the rate at which a body may release internal energy in the form of heat. In its simplest form, the Second Law of Thermodynamics expresses the fact

that each body has a limit on the rate at which heat can be absorbed, but has no limit on the rate at which heat can be released.

The Second Law has a long and complicated history. It has its origins in the study of homogeneous systems undergoing reversible processes, and its extension to the general framework of continuum mechanics is not entirely settled. Here we adopt a form of the Second Law called the Clausius–Duhem inequality which will suffice for our purposes. In particular, we introduce this inequality primarily to illustrate how thermodynamic considerations place restrictions on the constitutive relations of a body, and how various statements about dissipation and irreversibility may be deduced from it.

To motivate the form of the law given here, we first consider a homogeneous body Ω_t with uniform temperature $\Theta[\Omega_t]$ for each $t \geq 0$. For such a body the Second Law postulates the existence of an a-priori least upper bound $\Xi[\Omega_t]$ for the net heating $Q[\Omega_t]$ so that

$$Q[\Omega_t] \leq \Xi[\Omega_t]. \quad (5.16)$$

The quantity $\Xi[\Omega_t]$ in general depends on properties of the body Ω_t . When the net working is zero, we deduce from the First Law that $\frac{d}{dt}U[\Omega_t] = Q[\Omega_t]$. Thus the quantity $\Xi[\Omega_t]$ can also be interpreted as the maximum rate at which a body can store internal energy in the absence of net working.

By the **entropy** of a homogeneous body Ω_t we mean a quantity $\mathcal{H}[\Omega_t]$ defined, up to an additive constant, by

$$\frac{d}{dt}\mathcal{H}[\Omega_t] = \frac{\Xi[\Omega_t]}{\Theta[\Omega_t]}.$$

Thus the entropy of a body is a quantity whose rate of change at any instant of time is equal to the upper heating bound per unit temperature. In particular, the rate of change of entropy is a measure of a body's ability to absorb heat. At the atomic level, entropy can be interpreted as a measure of disorder, that is, the multitude of atomic configurations compatible with prescribed values of macroscopic variables.

In terms of entropy, the bound in (5.16) becomes

$$\frac{d}{dt}\mathcal{H}[\Omega_t] \geq \frac{Q[\Omega_t]}{\Theta[\Omega_t]}. \quad (5.17)$$

This is a classic form of the Second Law for homogeneous systems called the **Clausius–Planck inequality**. The irreversibility of natural pro-

cesses is reflected in the fact that

$$\frac{d}{dt} \mathcal{H}[\Omega_t] \geq 0 \quad \text{when} \quad Q[\Omega_t] = 0.$$

In particular, the entropy of a body favors increase, even when the net heating vanishes as in the case of an isolated body.

We next drop the assumption of homogeneity and consider an extension of the Clausius–Planck inequality to a general continuum body. Just as with internal energy, we assume the entropy $\mathcal{H}[\Omega_t]$ of a body is a continuous function of volume. In particular, we assume there exists an **entropy density field** per unit volume $\hat{\eta}(\mathbf{x}, t) \in \mathbb{R}$ such that

$$\mathcal{H}[\Omega_t] = \int_{\Omega_t} \hat{\eta}(\mathbf{x}, t) dV_{\mathbf{x}}.$$

For convenience, we introduce an entropy density per unit mass by

$$\eta(\mathbf{x}, t) = \rho^{-1}(\mathbf{x}, t) \hat{\eta}(\mathbf{x}, t),$$

so that

$$\mathcal{H}[\Omega_t] = \int_{\Omega_t} \rho(\mathbf{x}, t) \eta(\mathbf{x}, t) dV_{\mathbf{x}}.$$

A natural generalization of (5.17) to spatially inhomogeneous systems may be stated as follows.

Axiom 5.4 Second Law of Thermodynamics. *The rate of entropy production in any open subset of a continuum body is bounded below by the heating per unit temperature. That is*

$$\frac{d}{dt} \mathcal{H}[\Omega_t] \geq \int_{\Omega_t} \frac{\rho(\mathbf{x}, t) r(\mathbf{x}, t)}{\theta(\mathbf{x}, t)} dV_{\mathbf{x}} - \int_{\partial\Omega_t} \frac{\mathbf{q}(\mathbf{x}, t) \cdot \mathbf{n}(\mathbf{x})}{\theta(\mathbf{x}, t)} dA_{\mathbf{x}}, \quad \forall \Omega_t \subseteq B_t.$$

□

The above formulation of the Second Law is called the **Clausius–Duhem inequality**. We assume that it holds for every motion and every subset of a continuum body and study the consequences. As we shall see, the above inequality places certain restrictions on the constitutive relations of a body, and will lead to various statements about the dissipation of energy and flow of heat. While the precise form of the Second Law in continuum mechanics is not settled, we expect similar consequences from any other inequality which might replace Axiom 5.4.

5.2.3 Integral Versus Local Balance Laws

In the above developments, we have summarized balance laws for mass, momentum, energy and entropy in terms of arbitrary open subsets Ω_t of the current configuration B_t of a continuum body undergoing a motion $\varphi : B \times [0, \infty) \rightarrow \mathbf{E}^3$. We next exploit the Localization Theorem (see Result 2.19) to obtain local forms of these balance laws in terms of partial differential equations. When the local form of a balance law is formulated in terms of the current coordinates $\mathbf{x} \in B_t$ and time t , we say that it is in **spatial** or **Eulerian** form. When a balance law is formulated in terms of the reference coordinates $\mathbf{X} \in B$ and time t , we say that it is in **material** or **Lagrangian** form. In the next two sections we develop local statements of the balance laws in both Eulerian and Lagrangian forms.

5.3 Localized Eulerian Form of Balance Laws

Consider a continuum body with reference configuration B undergoing a motion $\varphi : B \times [0, \infty) \rightarrow \mathbf{E}^3$. As before, we denote the current configuration by $B_t = \varphi_t(B)$, and we assume $B_0 = B$ so that φ_0 is the identity (see Figure 4.7). For each $t \geq 0$ we assume the deformation $\varphi_t : B \rightarrow B_t$ is admissible, and we further assume $\varphi(\mathbf{X}, t)$ is smooth in the sense that partial derivatives of all orders exist and are continuous. This is much stronger than what we need to derive local statements, but it simplifies the exposition.

5.3.1 Conservation of Mass

Consider an arbitrary open subset Ω_t of B_t and let Ω be the corresponding subset in B so that $\Omega_t = \varphi_t(\Omega)$. Then Axiom 5.1 implies

$$\text{mass}[\Omega_t] = \text{mass}[\Omega_0],$$

where $\Omega_0 = \Omega$. In view of Result 4.8 we have

$$\text{mass}[\Omega_t] = \int_{\Omega_t} \rho(\mathbf{x}, t) dV_{\mathbf{x}} = \int_{\Omega} \rho_m(\mathbf{X}, t) \det \mathbf{F}(\mathbf{X}, t) dV_{\mathbf{X}},$$

where ρ_m is the material description of the spatial field ρ , that is, $\rho_m(\mathbf{X}, t) = \rho(\varphi(\mathbf{X}, t), t)$. Also, since $\varphi(\mathbf{X}, 0) = \mathbf{X}$, we have

$$\text{mass}[\Omega_0] = \int_{\Omega} \rho(\mathbf{X}, 0) dV_{\mathbf{X}} = \int_{\Omega} \rho_0(\mathbf{X}) dV_{\mathbf{X}},$$

where $\rho_0(\mathbf{X}) = \rho(\mathbf{X}, 0)$. Thus conservation of mass requires

$$\int_{\Omega} [\rho_m(\mathbf{X}, t) \det \mathbf{F}(\mathbf{X}, t) - \rho_0(\mathbf{X})] dV_{\mathbf{X}} = 0,$$

for any time $t \geq 0$ and open set Ω in B . By the Localization Theorem we deduce that

$$\rho_m(\mathbf{X}, t) \det \mathbf{F}(\mathbf{X}, t) = \rho_0(\mathbf{X}), \quad \forall \mathbf{X} \in B, t \geq 0. \quad (5.18)$$

As stated above, the law of conservation of mass is in the material or Lagrangian form. We next convert this to the spatial or Eulerian form. Taking the time derivative of both sides of (5.18) gives

$$\left\{ \frac{\partial}{\partial t} \rho(\varphi(\mathbf{X}, t), t) \right\} \det \mathbf{F}(\mathbf{X}, t) + \rho(\varphi(\mathbf{X}, t), t) \left\{ \frac{\partial}{\partial t} \det \mathbf{F}(\mathbf{X}, t) \right\} = 0.$$

Using Result 4.9 for the time-derivative of the Jacobian field $\det \mathbf{F}$ we obtain

$$\begin{aligned} & \left\{ \frac{\partial}{\partial t} \rho(\varphi(\mathbf{X}, t), t) \right\} \det \mathbf{F}(\mathbf{X}, t) \\ & + \rho(\varphi(\mathbf{X}, t), t) \det \mathbf{F}(\mathbf{X}, t) (\nabla^x \cdot \mathbf{v})(\mathbf{x}, t) \Big|_{\mathbf{x}=\varphi(\mathbf{X}, t)} = 0. \end{aligned}$$

Dividing through by $\det \mathbf{F}$, which is positive for any admissible motion, and using Result 4.7 for the total time derivative of a spatial field we deduce

$$\frac{\partial}{\partial t} \rho(\mathbf{x}, t) + \nabla^x \rho(\mathbf{x}, t) \cdot \mathbf{v}(\mathbf{x}, t) + \rho(\mathbf{x}, t) (\nabla^x \cdot \mathbf{v})(\mathbf{x}, t) = 0.$$

Finally, with the aid of the identity $\nabla^x \cdot (\phi \mathbf{w}) = \nabla^x \phi \cdot \mathbf{w} + \phi (\nabla^x \cdot \mathbf{w})$, which holds for an arbitrary scalar field ϕ and vector field $\mathbf{w}$, we obtain the following result.

Result 5.5 Conservation of Mass in Eulerian Form. *Let $\varphi : B \times [0, \infty) \rightarrow \mathbf{E}^3$ be a motion of a continuum body with associated spatial velocity field $\mathbf{v}(\mathbf{x}, t)$ and spatial mass density field $\rho(\mathbf{x}, t)$. Then conservation of mass requires*

$$\frac{\partial}{\partial t} \rho + \nabla^x \cdot (\rho \mathbf{v}) = 0, \quad \forall \mathbf{x} \in B_t, t \geq 0.$$

Equivalently, by definition of the total time derivative, we have

$$\dot{\rho} + \rho \nabla^x \cdot \mathbf{v} = 0, \quad \forall \mathbf{x} \in B_t, t \geq 0.$$

□

For a related result derived under different assumptions see Exercise 6. Because the current configuration B_t of a continuum body depends on time, any integral over B_t will also depend on time. We next exploit Result 5.5 to derive a simple expression for the time derivative of integrals over B_t expressed relative to the mass density ρ .

Result 5.6 Time Derivative of Integrals Relative to Mass. *Let $\varphi : B \times [0, \infty) \rightarrow \mathbf{E}^3$ be a motion of a continuum body with associated spatial velocity field $\mathbf{v}(\mathbf{x}, t)$ and spatial mass density field $\rho(\mathbf{x}, t)$. Let $\Phi(\mathbf{x}, t)$ be any spatial scalar, vector or second-order tensor field, and let Ω_t be an arbitrary open subset of B_t . Then*

$$\frac{d}{dt} \int_{\Omega_t} \Phi(\mathbf{x}, t) \rho(\mathbf{x}, t) dV_{\mathbf{x}} = \int_{\Omega_t} \dot{\Phi}(\mathbf{x}, t) \rho(\mathbf{x}, t) dV_{\mathbf{x}}.$$

□

Proof A slight generalization of Result 4.8 from scalar fields to vector and second-order tensor fields gives

$$\int_{\Omega_t} \Phi(\mathbf{x}, t) \rho(\mathbf{x}, t) dV_{\mathbf{x}} = \int_{\Omega} \Phi(\varphi(\mathbf{X}, t), t) \rho(\varphi(\mathbf{X}, t), t) \det \mathbf{F}(\mathbf{X}, t) dV_{\mathbf{X}},$$

where Ω is the corresponding subset of B , that is, $\Omega_t = \varphi_t(\Omega)$. Using conservation of mass in the form (5.18), we obtain

$$\int_{\Omega_t} \Phi(\mathbf{x}, t) \rho(\mathbf{x}, t) dV_{\mathbf{x}} = \int_{\Omega} \Phi(\varphi(\mathbf{X}, t), t) \rho_0(\mathbf{X}) dV_{\mathbf{X}}.$$

Noting that the region Ω in B does not depend on time, we take time derivatives and obtain

$$\begin{aligned} \frac{d}{dt} \int_{\Omega_t} \Phi(\mathbf{x}, t) \rho(\mathbf{x}, t) dV_{\mathbf{x}} &= \frac{d}{dt} \int_{\Omega} \Phi(\varphi(\mathbf{X}, t), t) \rho_0(\mathbf{X}) dV_{\mathbf{X}} \\ &= \int_{\Omega} \frac{\partial}{\partial t} \Phi(\varphi(\mathbf{X}, t), t) \rho_0(\mathbf{X}) dV_{\mathbf{X}} \\ &= \int_{\Omega} \left\{ \frac{\partial}{\partial t} \Phi(\varphi(\mathbf{X}, t), t) \right\} \rho(\varphi(\mathbf{X}, t), t) \det \mathbf{F}(\mathbf{X}, t) dV_{\mathbf{X}}, \end{aligned}$$

where the last line follows from (5.18). Using the relationship

$$\dot{\Phi}(\mathbf{x}, t) \Big|_{\mathbf{x}=\varphi(\mathbf{X}, t)} = \frac{\partial}{\partial t} \Phi(\varphi(\mathbf{X}, t), t),$$

together with Result 4.8, we get

$$\begin{aligned}
 \frac{d}{dt} \int_{\Omega_t} \Phi(\mathbf{x}, t) \rho(\mathbf{x}, t) dV_{\mathbf{x}} \\
 &= \int_{\Omega} \dot{\Phi}(\mathbf{x}, t) \Big|_{\mathbf{x}=\boldsymbol{\varphi}(\mathbf{X}, t)} \rho(\boldsymbol{\varphi}(\mathbf{X}, t), t) \det \mathbf{F}(\mathbf{X}, t) dV_{\mathbf{X}} \\
 &= \int_{\Omega_t} \dot{\Phi}(\mathbf{x}, t) \rho(\mathbf{x}, t) dV_{\mathbf{x}},
 \end{aligned}$$

which is the desired result. $\square$

5.3.2 Balance of Linear Momentum

As stated in Axiom 5.2, the balance law for linear momentum for an arbitrary open subset Ω_t of B_t is

$$\begin{aligned}
 \frac{d}{dt} \int_{\Omega_t} \rho(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t) dV_{\mathbf{x}} \\
 = \int_{\partial\Omega_t} \mathbf{t}(\mathbf{x}, t) dA_{\mathbf{x}} + \int_{\Omega_t} \rho(\mathbf{x}, t) \mathbf{b}(\mathbf{x}, t) dV_{\mathbf{x}},
 \end{aligned} \tag{5.19}$$

where $\rho(\mathbf{x}, t)$ is the spatial mass density field, $\mathbf{v}(\mathbf{x}, t)$ is the spatial velocity field, $\mathbf{b}(\mathbf{x}, t)$ is the spatial body force field per unit mass and $\mathbf{t}(\mathbf{x}, t)$ is the traction field on the surface $\partial\Omega_t$ in B_t .

From Chapter 3 we recall that the traction field on any material surface in the current configuration B_t is determined by the Cauchy stress field $\mathbf{S}(\mathbf{x}, t)$ as

$$\mathbf{t}(\mathbf{x}, t) = \mathbf{S}(\mathbf{x}, t) \mathbf{n}(\mathbf{x}),$$

where $\mathbf{n}(\mathbf{x})$ is the outward unit normal field on the surface. Thus we can rewrite (5.19) as (omitting the arguments $\mathbf{x}$ and t for clarity)

$$\frac{d}{dt} \int_{\Omega_t} \rho \mathbf{v} dV_{\mathbf{x}} = \int_{\partial\Omega_t} \mathbf{S} \mathbf{n} dA_{\mathbf{x}} + \int_{\Omega_t} \rho \mathbf{b} dV_{\mathbf{x}}.$$

Using Result 5.6 and the Divergence Theorem for second-order tensor fields we obtain

$$\int_{\Omega_t} \rho \dot{\mathbf{v}} dV_{\mathbf{x}} = \int_{\Omega_t} [\nabla^x \cdot \mathbf{S} + \rho \mathbf{b}] dV_{\mathbf{x}}.$$

The fact that this must hold for an arbitrary open subset Ω_t of B_t , together with the Localization Theorem, leads to the following result.

Result 5.7 Law of Linear Momentum in Eulerian Form. *Let $\varphi : B \times [0, \infty) \rightarrow \mathbf{E}^3$ be a motion of a continuum body with associated spatial velocity field $\mathbf{v}(\mathbf{x}, t)$ and spatial mass density field $\rho(\mathbf{x}, t)$. Then balance of linear momentum requires*

$$\rho \dot{\mathbf{v}} = \nabla^x \cdot \mathbf{S} + \rho \mathbf{b}, \quad \forall \mathbf{x} \in B_t, \quad t \geq 0,$$

where $\mathbf{S}(\mathbf{x}, t)$ is the Cauchy stress field and $\mathbf{b}(\mathbf{x}, t)$ is the spatial body force per unit mass. □

The above equations may be viewed as a generalization to the dynamic case of the equilibrium (balance of forces) condition described in Result 3.5. In particular, the condition of equilibrium corresponds to the special case when $\dot{\mathbf{v}} = \mathbf{0}$ for all $\mathbf{x} \in B_t$ and $t \geq 0$.

Remarks:

- (1) By Result 4.7 we have

$$\dot{\mathbf{v}} = \frac{\partial}{\partial t} \mathbf{v} + (\nabla^x \mathbf{v}) \mathbf{v}.$$

Thus the balance of linear momentum equation may be written in the equivalent form

$$\rho \left[\frac{\partial}{\partial t} \mathbf{v} + (\nabla^x \mathbf{v}) \mathbf{v} \right] = \nabla^x \cdot \mathbf{S} + \rho \mathbf{b}.$$

- (2) As mentioned in Chapter 4, many texts use the notation $\mathbf{v} \cdot \nabla^x \mathbf{v}$ in place of $(\nabla^x \mathbf{v}) \mathbf{v}$. We, however, will always use $(\nabla^x \mathbf{v}) \mathbf{v}$, which denotes the application of the second-order tensor $\nabla^x \mathbf{v}$ to the vector $\mathbf{v}$. □

5.3.3 Balance of Angular Momentum

As stated in Axiom 5.2, the balance law for angular momentum (about the origin) for an arbitrary open subset Ω_t of B_t is

$$\begin{aligned} & \frac{d}{dt} \int_{\Omega_t} \mathbf{x} \times \rho(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t) \, dV_{\mathbf{x}} \\ &= \int_{\partial \Omega_t} \mathbf{x} \times \mathbf{t}(\mathbf{x}, t) \, dA_{\mathbf{x}} + \int_{\Omega_t} \mathbf{x} \times \rho(\mathbf{x}, t) \mathbf{b}(\mathbf{x}, t) \, dV_{\mathbf{x}}, \end{aligned} \tag{5.20}$$

where $\rho(\mathbf{x}, t)$ is the spatial mass density field, $\mathbf{v}(\mathbf{x}, t)$ is the spatial velocity field, $\mathbf{b}(\mathbf{x}, t)$ is the spatial body force field per unit mass and $\mathbf{t}(\mathbf{x}, t)$ is the traction field on the surface $\partial\Omega_t$ in B_t .

To simplify the left-hand side of (5.20) we notice that the relation $\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X}, t)$ implies $\dot{\mathbf{x}} = \mathbf{v}(\mathbf{x}, t)$. Thus

$$\frac{d}{dt}[\mathbf{x} \times \mathbf{v}(\mathbf{x}, t)] = \dot{\mathbf{x}} \times \mathbf{v}(\mathbf{x}, t) + \mathbf{x} \times \dot{\mathbf{v}}(\mathbf{x}, t) = \mathbf{x} \times \dot{\mathbf{v}}(\mathbf{x}, t),$$

and by Result 5.6 we get

$$\begin{aligned} \frac{d}{dt} \int_{\Omega_t} \rho(\mathbf{x}, t) [\mathbf{x} \times \mathbf{v}(\mathbf{x}, t)] dV_{\mathbf{x}} &= \int_{\Omega_t} \rho(\mathbf{x}, t) \frac{d}{dt} [\mathbf{x} \times \mathbf{v}(\mathbf{x}, t)] dV_{\mathbf{x}} \\ &= \int_{\Omega_t} \rho(\mathbf{x}, t) [\mathbf{x} \times \dot{\mathbf{v}}(\mathbf{x}, t)] dV_{\mathbf{x}}. \end{aligned}$$

Substituting the above result into (5.20) and using the definition of the Cauchy stress field $\mathbf{S}(\mathbf{x}, t)$ we obtain

$$\begin{aligned} &\int_{\Omega_t} \rho(\mathbf{x}, t) [\mathbf{x} \times \dot{\mathbf{v}}(\mathbf{x}, t)] dV_{\mathbf{x}} \\ &= \int_{\partial\Omega_t} \mathbf{x} \times \mathbf{S}(\mathbf{x}, t) \mathbf{n} dA_{\mathbf{x}} + \int_{\Omega_t} \rho(\mathbf{x}, t) [\mathbf{x} \times \mathbf{b}(\mathbf{x}, t)] dV_{\mathbf{x}}. \end{aligned}$$

By employing Result 5.7 the above expression may be reduced to

$$\int_{\partial\Omega_t} \mathbf{x} \times \mathbf{S}(\mathbf{x}, t) \mathbf{n} dA_{\mathbf{x}} - \int_{\Omega_t} \mathbf{x} \times (\nabla^x \cdot \mathbf{S})(\mathbf{x}, t) dV_{\mathbf{x}} = \mathbf{0},$$

which must hold for an arbitrary open subset Ω_t of B_t . Proceeding as in the proof of Result 3.5 we arrive at the following result.

Result 5.8 Law of Angular Momentum in Eulerian Form. *Let $\boldsymbol{\varphi} : B \times [0, \infty) \rightarrow \mathbf{E}^3$ be a motion of a continuum body and let $\mathbf{S}(\mathbf{x}, t)$ denote the Cauchy stress field in B_t . Then balance of angular momentum requires*

$$\mathbf{S}^T = \mathbf{S}, \quad \forall \mathbf{x} \in B_t, t \geq 0.$$

□

Thus balance of angular momentum requires that the Cauchy stress field be symmetric at each point and time in the motion of a body. Notice that this result is identical to the one obtained in Result 3.5 of Chapter 3, which was based only on equilibrium arguments. Thus the Cauchy stress field must be symmetric whether a body is in motion or

in equilibrium. For a related result derived under different assumptions see Exercise 7.

5.3.4 Characterization of Net Working

Before proceeding to study localized versions of the First and Second Laws of Thermodynamics, we use Results 5.7 and 5.8 to derive a relation between the rate of change of kinetic energy and the power of external and internal forces. As we will see, this relation will lead to an explicit expression for the net working on a body.

To begin, consider Result 5.7 and take the dot product with the spatial velocity field $\mathbf{v}(\mathbf{x}, t)$ to obtain

$$\rho \mathbf{v} \cdot \dot{\mathbf{v}} = (\nabla^x \cdot \mathbf{S}) \cdot \mathbf{v} + \rho \mathbf{b} \cdot \mathbf{v}, \quad (5.21)$$

for all $\mathbf{x} \in B_t$ and $t \geq 0$. Integration of (5.21) over an arbitrary open subset Ω_t of B_t yields

$$\int_{\Omega_t} \rho \mathbf{v} \cdot \dot{\mathbf{v}} dV_{\mathbf{x}} = \int_{\Omega_t} [(\nabla^x \cdot \mathbf{S}) \cdot \mathbf{v} + \rho \mathbf{b} \cdot \mathbf{v}] dV_{\mathbf{x}}.$$

Using the identity in Result 2.11, namely

$$\nabla^x \cdot (\mathbf{S}^T \mathbf{v}) = (\nabla^x \cdot \mathbf{S}) \cdot \mathbf{v} + \mathbf{S} : \nabla^x \mathbf{v},$$

together with the Divergence Theorem, we obtain

$$\begin{aligned} & \int_{\Omega_t} \rho \mathbf{v} \cdot \dot{\mathbf{v}} dV_{\mathbf{x}} \\ &= \int_{\Omega_t} -\mathbf{S} : \nabla^x \mathbf{v} dV_{\mathbf{x}} + \int_{\partial\Omega_t} \mathbf{S}^T \mathbf{v} \cdot \mathbf{n} dA_{\mathbf{x}} + \int_{\Omega_t} \rho \mathbf{b} \cdot \mathbf{v} dV_{\mathbf{x}}. \end{aligned}$$

From Results 5.8 and 1.13 we deduce $\mathbf{S} : \nabla^x \mathbf{v} = \mathbf{S} : \mathbf{L}$, where $\mathbf{L} = \text{sym}(\nabla^x \mathbf{v})$ is the rate of strain field. Substitution of this result into the above expression yields

$$\int_{\Omega_t} \rho \mathbf{v} \cdot \dot{\mathbf{v}} dV_{\mathbf{x}} + \int_{\Omega_t} \mathbf{S} : \mathbf{L} dV_{\mathbf{x}} = \int_{\partial\Omega_t} \mathbf{v} \cdot \mathbf{S} \mathbf{n} dA_{\mathbf{x}} + \int_{\Omega_t} \rho \mathbf{b} \cdot \mathbf{v} dV_{\mathbf{x}},$$

which leads to the following result.

Result 5.9 Net Working in Eulerian Form. *Let $\varphi : B \times [0, \infty) \rightarrow \mathbb{E}^3$ be a motion of a continuum body with associated Cauchy stress field $\mathbf{S}(\mathbf{x}, t)$ and rate of strain field $\mathbf{L}(\mathbf{x}, t)$, and let Ω_t be an arbitrary open*

subset of B_t . Then

$$\frac{d}{dt}K[\Omega_t] + \int_{\Omega_t} \mathbf{S} : \mathbf{L} \, dV_{\mathbf{x}} = \mathcal{P}[\Omega_t], \quad \forall t \geq 0,$$

where $K[\Omega_t]$ is the total kinetic energy and $\mathcal{P}[\Omega_t]$ is the power of external forces on Ω_t . Thus, in view of (5.14), the net working $\mathcal{W}[\Omega_t]$ is given by

$$\mathcal{W}[\Omega_t] = \int_{\Omega_t} \mathbf{S} : \mathbf{L} \, dV_{\mathbf{x}}, \quad \forall t \geq 0.$$

□

The quantity $\mathbf{S} : \mathbf{L}$ is typically called the **stress power** associated with a motion. It corresponds to the rate of work done by internal forces (stresses) at each point in a continuum body. The above result shows that the net working on any open subset of a body is equal to the integral of the stress power over that subset.

5.3.5 First Law of Thermodynamics

As stated in Axiom 5.3, the balance law of energy for an arbitrary open subset Ω_t of B_t is

$$\frac{d}{dt} \int_{\Omega_t} \rho \phi \, dV_{\mathbf{x}} = \int_{\Omega_t} \mathbf{S} : \mathbf{L} \, dV_{\mathbf{x}} - \int_{\partial\Omega_t} \mathbf{q} \cdot \mathbf{n} \, dA_{\mathbf{x}} + \int_{\Omega_t} \rho r \, dV_{\mathbf{x}},$$

where the expressions in Result 5.9 and equation (5.13) for the net working and heating, respectively, have been used. Here $\rho(\mathbf{x}, t)$ is the mass density field, $\phi(\mathbf{x}, t)$ is the internal energy field per unit mass, $\mathbf{q}(\mathbf{x}, t)$ is the Fourier–Stokes heat flux vector field, $r(\mathbf{x}, t)$ is the heat supply field per unit mass, $\mathbf{L}(\mathbf{x}, t)$ is the rate of strain field and $\mathbf{S}(\mathbf{x}, t)$ is the Cauchy stress field in B_t .

Using Result 5.6 together with the Divergence Theorem we obtain

$$\int_{\Omega_t} \rho \dot{\phi} \, dV_{\mathbf{x}} = \int_{\Omega_t} \mathbf{S} : \mathbf{L} \, dV_{\mathbf{x}} - \int_{\Omega_t} \nabla^x \cdot \mathbf{q} \, dV_{\mathbf{x}} + \int_{\Omega_t} \rho r \, dV_{\mathbf{x}}.$$

The fact that the above statement must hold for an arbitrary open subset Ω_t of B_t , together with the Localization Theorem, leads to the following result.

Result 5.10 Law of Energy in Eulerian Form. Let $\varphi : B \times [0, \infty) \rightarrow \mathbf{E}^3$ be a motion of a continuum body. Then

$$\rho \dot{\phi} = \mathbf{S} : \mathbf{L} - \nabla^x \cdot \mathbf{q} + \rho r, \quad \forall \mathbf{x} \in B_t, t \geq 0.$$

□

5.3.6 Second Law of Thermodynamics

As stated in Axiom 5.4, the Clausius–Duhem form of the Second Law of Thermodynamics for an arbitrary open subset Ω_t of B_t is

$$\frac{d}{dt} \int_{\Omega_t} \rho \eta \, dV_{\mathbf{x}} \geq \int_{\Omega_t} \frac{\rho r}{\theta} \, dV_{\mathbf{x}} - \int_{\partial \Omega_t} \frac{\mathbf{q} \cdot \mathbf{n}}{\theta} \, dA_{\mathbf{x}},$$

where $\rho(\mathbf{x}, t)$ is the mass density field, $\eta(\mathbf{x}, t)$ is the entropy field per unit mass, $\mathbf{q}(\mathbf{x}, t)$ is the Fourier–Stokes heat flux vector field, $r(\mathbf{x}, t)$ is the heat supply field per unit mass and $\theta(\mathbf{x}, t)$ is the (absolute) temperature field in B_t . An application of the Divergence Theorem followed by the Localization Theorem leads to the following local statement.

Result 5.11 Clausius–Duhem Inequality in Eulerian Form. Let $\varphi : B \times [0, \infty) \rightarrow \mathbf{E}^3$ be a motion of a continuum body. Then

$$\rho \dot{\eta} \geq \theta^{-1} \rho r - \nabla^x \cdot (\theta^{-1} \mathbf{q}), \quad \forall \mathbf{x} \in B_t, t \geq 0. \quad (5.22)$$

□

By expanding the divergence term in (5.22) and multiplying through by θ we obtain

$$\theta \rho \dot{\eta} \geq \rho r - \nabla^x \cdot \mathbf{q} + \theta^{-1} \mathbf{q} \cdot \nabla^x \theta. \quad (5.23)$$

The above expression can be written in the equivalent form

$$\delta - \theta^{-1} \mathbf{q} \cdot \nabla^x \theta \geq 0, \quad (5.24)$$

where $\delta = \theta \rho \dot{\eta} - (\rho r - \nabla^x \cdot \mathbf{q})$ is called the **internal dissipation density field** per unit volume. It represents the difference between the local rate of entropy increase and the local heating. In other words, δ is a measure of the local rate of entropy increase due to mechanisms other than the local supply and transfer of heat.

Various statements about the internal dissipation δ and the heat flux $\mathbf{q}$ can be made on the basis of (5.24). For example, at any point at which the temperature satisfies $\nabla^x \theta = \mathbf{0}$, the internal dissipation must

satisfy $\delta \geq 0$. Thus, the internal dissipation is everywhere non-negative in bodies with spatially homogeneous temperature. From (5.24) we also deduce that, at any point at which $\delta = 0$, the heat flux and temperature must satisfy $\mathbf{q} \cdot \nabla^x \theta \leq 0$. Thus, the heat flux vector must everywhere make an obtuse angle with the temperature gradient in bodies with no internal dissipation. In particular, heat must flow from “hot” to “cold” in such bodies.

To study the consequences of the Clausius–Duhem inequality for various constitutive models introduced later, it is convenient to introduce the quantity

$$\psi(\mathbf{x}, t) = \phi(\mathbf{x}, t) - \theta(\mathbf{x}, t)\eta(\mathbf{x}, t), \quad (5.25)$$

which is called the **free energy density field** per unit mass. According to the classic theory of homogeneous systems undergoing reversible processes, free energy is that portion of the internal energy available for performing work at constant temperature. Result 5.11 can be restated in terms free energy as follows.

Result 5.12 *Reduced Clausius–Duhem Inequality in Eulerian Form.* *Let $\varphi : B \times [0, \infty) \rightarrow \mathbf{E}^3$ be a motion of a continuum body with spatial free energy field $\psi(\mathbf{x}, t)$. Then*

$$\rho \dot{\psi} \leq \mathbf{S} : \mathbf{L} - \rho \eta \dot{\theta} - \theta^{-1} \mathbf{q} \cdot \nabla^x \theta, \quad \forall \mathbf{x} \in B_t, t \geq 0. \quad (5.26)$$

□

Proof From (5.25) we obtain $\theta \dot{\eta} = \dot{\phi} - \dot{\theta} \eta - \dot{\psi}$ (see Exercise 8). Multiplying this expression by ρ and using Result 5.10 to eliminate $\rho \dot{\phi}$ we get

$$\rho \theta \dot{\eta} = \mathbf{S} : \mathbf{L} - \nabla^x \cdot \mathbf{q} + \rho r - \rho \dot{\theta} \eta - \rho \dot{\psi}.$$

Substituting this expression into (5.23) leads to the desired result. □

Various statements about the free energy can be deduced from Result 5.12. For example, in bodies with constant, spatially homogeneous temperature we must have $\rho \dot{\psi} \leq \mathbf{S} : \mathbf{L}$. If we further assume that a body can only undergo reversible processes in the sense that equality is always achieved in the Clausius–Duhem inequality, we obtain $\rho \dot{\psi} = \mathbf{S} : \mathbf{L}$. Thus for such bodies the rate of change of free energy is equal to the stress power. The term “reduced” in Result 5.12 reflects the fact that, when considered within the classic case of homogeneous bodies, the inequality

in (5.26) is independent of the heat supply r and heat flux $\mathbf{q}$ in contrast to (5.22).

5.3.7 Summary

In the Eulerian description of the motion of a general continuum body there are 22 basic unknown fields:

$\varphi_i(\mathbf{X}, t)$	3 components of motion
$v_i(\mathbf{x}, t)$	3 components of velocity
$\rho(\mathbf{x}, t)$	1 mass density
$S_{ij}(\mathbf{x}, t)$	9 components of stress
$\theta(\mathbf{x}, t)$	1 temperature
$q_i(\mathbf{x}, t)$	3 components of heat flux
$\phi(\mathbf{x}, t)$	1 internal energy per unit mass
$\eta(\mathbf{x}, t)$	1 entropy per unit mass.

To determine these unknown fields we have the following 11 equations:

$[v_i]_m = \frac{\partial}{\partial t} \varphi_i$	3 kinematical
$\frac{\partial}{\partial t} \rho + (\rho v_i)_{,i} = 0$	1 mass
$\rho \dot{v}_i = S_{ij,j} + \rho b_i$	3 linear momentum
$S_{ij} = S_{ji}$	3 independent angular momentum
$\rho \dot{\phi} = S_{ij} v_{i,j} - q_{i,i} + \rho r$	1 energy.

Remarks:

- (1) The motion map φ is often not needed within the Eulerian framework. In particular, when the current placement B_t is known or specified for all $t \geq 0$, the balance laws can typically be applied without explicit knowledge of φ . The number of unknowns is then reduced to 19 (φ disappears), and the number of equations is reduced to 8 (kinematical equations disappear).
- (2) Since the number of unknowns is greater than the number of equations by 11, extra equations are required to close the system. As we will see later, system closure can be obtained by the introduction of **constitutive equations** which relate $(\mathbf{S}, \mathbf{q}, \phi, \eta)$ to

$(\rho, \mathbf{v}, \theta)$. Such relations reflect the specific material properties of a body (see Exercise 9).

- (3) The Second Law of Thermodynamics (Clausius–Duhem inequality) does not provide an equation with which to determine unknown fields. Instead, we interpret the Second Law as providing a restriction on constitutive relations. In particular, the constitutive relations of a body must guarantee that the Second Law is satisfied in every possible motion of that body.
- (4) When thermal effects are neglected, the number of unknowns is further reduced from 19 to 13 ($\mathbf{q}$, ϕ , η and θ disappear), and the number of equations is reduced from 8 to 7 (balance of energy disappears). In this case, system closure can be obtained by 6 constitutive equations which relate $\mathbf{S}$ to $(\rho, \mathbf{v})$. $\square$

5.4 Localized Lagrangian Form of Balance Laws

Again we consider a continuum body with reference configuration B undergoing a motion $\varphi : B \times [0, \infty) \rightarrow \mathbf{E}^3$. We denote the current configuration by $B_t = \varphi_t(B)$, and we assume $B_0 = B$ so that φ_0 is the identity. Since the deformation $\varphi_t : B \rightarrow B_t$ is assumed to be admissible for each $t \geq 0$, we have a bijection between B and B_t for each $t \geq 0$. Thus, by a change of variable, any balance law stated in terms of $\mathbf{x} \in B_t$ can be expressed in terms of $\mathbf{X} \in B$. This leads to the material or Lagrangian form of the balance laws.

5.4.1 Conservation of Mass

As before, consider an arbitrary open subset Ω_t of B_t , and let Ω be the corresponding subset in B so that $\Omega_t = \varphi_t(\Omega)$. Then Axiom 5.1 implies

$$\text{mass}[\Omega_t] = \text{mass}[\Omega_0],$$

where $\Omega_0 = \Omega$. In view of the arguments leading to (5.18) we have the following result.

Result 5.13 Conservation of Mass in Lagrangian Form. *Let $\varphi : B \times [0, \infty) \rightarrow \mathbf{E}^3$ be a motion of a continuum body with associated deformation gradient field $\mathbf{F}(\mathbf{X}, t)$ and spatial mass density field $\rho(\mathbf{x}, t)$.*

Let $\rho_0(\mathbf{X})$ be the mass density field in the reference configuration B . Then conservation of mass requires

$$\rho_m(\mathbf{X}, t) \det \mathbf{F}(\mathbf{X}, t) = \rho_0(\mathbf{X}), \quad \forall \mathbf{X} \in B, t \geq 0,$$

where $\rho_m(\mathbf{X}, t)$ is the material description of the spatial field $\rho(\mathbf{x}, t)$. $\square$

5.4.2 Balance of Linear Momentum

There are various ways to develop the Lagrangian form of the balance law for linear momentum. We may proceed directly from the integral form of the law in Axiom 5.2. Alternatively, we may integrate the local Eulerian form in Result 5.7 over an arbitrary open subset Ω_t of B_t , change variables to obtain an integral over an open subset Ω of B , and then localize. Here we follow the first approach.

As stated in Axiom 5.2, the balance law for linear momentum for an arbitrary open subset Ω_t of B_t is

$$\frac{d}{dt} \mathbf{l}[\Omega_t] = \mathbf{r}[\Omega_t], \quad (5.27)$$

where $\mathbf{l}[\Omega_t]$ is the linear momentum of Ω_t defined by

$$\mathbf{l}[\Omega_t] = \int_{\Omega_t} \rho(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t) dV_{\mathbf{x}},$$

and $\mathbf{r}[\Omega_t]$ is the resultant external force on Ω_t defined by

$$\mathbf{r}[\Omega_t] = \int_{\Omega_t} \rho(\mathbf{x}, t) \mathbf{b}(\mathbf{x}, t) dV_{\mathbf{x}} + \int_{\partial\Omega_t} \mathbf{S}(\mathbf{x}, t) \mathbf{n}(\mathbf{x}) dA_{\mathbf{x}}.$$

Performing a change of variable in the integral for $\mathbf{l}[\Omega_t]$ we obtain

$$\begin{aligned} \mathbf{l}[\Omega_t] &= \int_{\Omega_t} \rho(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t) dV_{\mathbf{x}} \\ &= \int_{\Omega} \rho(\boldsymbol{\varphi}(\mathbf{X}, t), t) \mathbf{v}(\boldsymbol{\varphi}(\mathbf{X}, t), t) \det \mathbf{F}(\mathbf{X}, t) dV_{\mathbf{X}} \\ &= \int_{\Omega} \rho_0(\mathbf{X}) \dot{\boldsymbol{\varphi}}(\mathbf{X}, t) dV_{\mathbf{X}}, \end{aligned} \quad (5.28)$$

where we have used Result 5.13. Performing a similar change of variable

in the integrals for $\mathbf{r}[\Omega_t]$ we obtain

$$\begin{aligned}\mathbf{r}[\Omega_t] &= \int_{\partial\Omega_t} \mathbf{S}(\mathbf{x}, t) \mathbf{n}(\mathbf{x}) \, dA_{\mathbf{x}} + \int_{\Omega_t} \rho(\mathbf{x}, t) \mathbf{b}(\mathbf{x}, t) \, dV_{\mathbf{x}} \\ &= \int_{\partial\Omega} (\det \mathbf{F}(\mathbf{X}, t)) \mathbf{S}(\boldsymbol{\varphi}(\mathbf{X}, t), t) \mathbf{F}(\mathbf{X}, t)^{-T} \mathbf{N}(\mathbf{X}) \, dA_{\mathbf{X}} \\ &\quad + \int_{\Omega} \det \mathbf{F}(\mathbf{X}, t) \rho(\boldsymbol{\varphi}(\mathbf{X}, t), t) \mathbf{b}(\boldsymbol{\varphi}(\mathbf{X}, t), t) \, dV_{\mathbf{X}},\end{aligned}$$

where we have used Result 4.11 for the transformation of the surface integral. In particular, $\mathbf{n}(\mathbf{x})$ is the unit outward normal field on $\partial\Omega_t$ and $\mathbf{N}(\mathbf{X})$ is the unit outward normal field on $\partial\Omega$. The following definition will help simplify our developments.

Definition 5.14 *Let $\mathbf{S}(\mathbf{x}, t)$ be the Cauchy stress field in B_t associated with a motion $\boldsymbol{\varphi} : B \times [0, \infty) \rightarrow \mathbf{E}^3$. Then by the **nominal** or **first Piola–Kirchhoff stress field** associated with $\boldsymbol{\varphi}$ we mean*

$$\mathbf{P}(\mathbf{X}, t) = (\det \mathbf{F}(\mathbf{X}, t)) \mathbf{S}_m(\mathbf{X}, t) \mathbf{F}(\mathbf{X}, t)^{-T}, \quad \forall \mathbf{X} \in B, t \geq 0,$$

*and by the **second Piola–Kirchhoff stress field** associated with $\boldsymbol{\varphi}$ we mean*

$$\boldsymbol{\Sigma}(\mathbf{X}, t) = \mathbf{F}(\mathbf{X}, t)^{-1} \mathbf{P}(\mathbf{X}, t), \quad \forall \mathbf{X} \in B, t \geq 0.$$

□

To proceed, let $\mathbf{b}_m$ be the material description of the spatial body force field $\mathbf{b}$. Then, using Result 5.13 and Definition 5.14, we find

$$\mathbf{r}[\Omega_t] = \int_{\partial\Omega} \mathbf{P}(\mathbf{X}, t) \mathbf{N}(\mathbf{X}) \, dA_{\mathbf{X}} + \int_{\Omega} \rho_0(\mathbf{X}) \mathbf{b}_m(\mathbf{X}, t) \, dV_{\mathbf{X}}. \quad (5.29)$$

Substituting (5.28) and (5.29) into (5.27) we obtain

$$\begin{aligned}\frac{d}{dt} \int_{\Omega} \rho_0(\mathbf{X}) \dot{\boldsymbol{\varphi}}(\mathbf{X}, t) \, dV_{\mathbf{X}} \\ = \int_{\partial\Omega} \mathbf{P}(\mathbf{X}, t) \mathbf{N}(\mathbf{X}) \, dA_{\mathbf{X}} + \int_{\Omega} \rho_0(\mathbf{X}) \mathbf{b}_m(\mathbf{X}, t) \, dV_{\mathbf{X}}.\end{aligned}$$

By the Divergence Theorem and the fact that Ω is independent of time we get

$$\begin{aligned}\int_{\Omega} \rho_0(\mathbf{X}) \ddot{\boldsymbol{\varphi}}(\mathbf{X}, t) \, dV_{\mathbf{X}} \\ = \int_{\Omega} [(\nabla^x \cdot \mathbf{P})(\mathbf{X}, t) + \rho_0(\mathbf{X}) \mathbf{b}_m(\mathbf{X}, t)] \, dV_{\mathbf{X}}.\end{aligned}$$

Because we have a bijection between B and B_t , any balance law which holds for an arbitrary subset Ω_t of B_t must also hold for an arbitrary subset Ω of B . This fact leads to the following result.

Result 5.15 Law of Linear Momentum in Lagrangian Form.

Let $\varphi : B \times [0, \infty) \rightarrow \mathbb{E}^3$ be a motion of a continuum body and let $\rho_0(\mathbf{X})$ denote the mass density field in the reference configuration B . Then balance of linear momentum requires

$$\rho_0 \ddot{\varphi} = \nabla^{\mathbf{x}} \cdot \mathbf{P} + \rho_0 \mathbf{b}_m, \quad \forall \mathbf{X} \in B, t \geq 0,$$

where $\mathbf{P}(\mathbf{X}, t)$ is the first Piola–Kirchhoff stress field and $\mathbf{b}_m(\mathbf{X}, t)$ is the material description of the spatial body force field $\mathbf{b}(\mathbf{x}, t)$. $\square$

5.4.3 Balance of Angular Momentum

The following is a straightforward consequence of Result 5.8 and Definition 5.14.

Result 5.16 Law of Angular Momentum in Lagrangian Form.

Let $\varphi : B \times [0, \infty) \rightarrow \mathbb{E}^3$ be a motion of a continuum body with associated deformation gradient field $\mathbf{F}(\mathbf{X}, t)$. Then balance of angular momentum requires

$$\mathbf{P}\mathbf{F}^T = \mathbf{F}\mathbf{P}^T \quad \text{or equivalently} \quad \boldsymbol{\Sigma}^T = \boldsymbol{\Sigma}, \quad \forall \mathbf{X} \in B, t \geq 0,$$

where $\mathbf{P}(\mathbf{X}, t)$ is the first and $\boldsymbol{\Sigma}(\mathbf{X}, t)$ is the second Piola–Kirchhoff stress field associated with φ . $\square$

5.4.4 Characterization of Net Working

Before proceeding to study localized Lagrangian versions of the First and Second Laws of Thermodynamics, we use Result 5.15 to derive a relation between the rate of change of kinetic energy and the power of external and internal forces. As before, this will lead to an explicit expression for the net working on a body.

To begin, consider an arbitrary open subset Ω_t of B_t , and let Ω be the corresponding subset in B . Then the kinetic energy of Ω_t is defined as

$$K[\Omega_t] = \int_{\Omega_t} \frac{1}{2} \rho(\mathbf{x}, t) |\mathbf{v}(\mathbf{x}, t)|^2 dV_{\mathbf{x}},$$

and the power of external forces on Ω_t is defined as

$$\mathcal{P}[\Omega_t] = \int_{\Omega_t} \rho(\mathbf{x}, t) \mathbf{b}(\mathbf{x}, t) \cdot \mathbf{v}(\mathbf{x}, t) dV_{\mathbf{x}} + \int_{\partial\Omega_t} \mathbf{S}(\mathbf{x}, t)^T \mathbf{v}(\mathbf{x}, t) \cdot \mathbf{n}(\mathbf{x}) dA_{\mathbf{x}}.$$

Notice that in the expression for $\mathcal{P}[\Omega_t]$ we have made use of the identity $\mathbf{S}^T \mathbf{v} \cdot \mathbf{n} = \mathbf{v} \cdot \mathbf{S} \mathbf{n}$.

Performing a change of variable in the integral for $K[\Omega_t]$ and using Result 5.13 we obtain

$$K[\Omega_t] = \int_{\Omega} \frac{1}{2} \rho_0(\mathbf{X}) |\dot{\boldsymbol{\varphi}}(\mathbf{X}, t)|^2 dV_{\mathbf{X}}. \quad (5.30)$$

Similarly, performing a change of variable in the integrals for $\mathcal{P}[\Omega_t]$ we obtain (omitting the arguments $\mathbf{X}$ and t for brevity)

$$\begin{aligned} \mathcal{P}[\Omega_t] &= \int_{\Omega} \rho_0 \mathbf{b}_m \cdot \mathbf{v}_m dV_{\mathbf{X}} + \int_{\partial\Omega} (\det \mathbf{F}) \mathbf{S}_m^T \mathbf{v}_m \cdot \mathbf{F}^{-T} \mathbf{N} dA_{\mathbf{X}} \\ &= \int_{\Omega} \rho_0 \mathbf{b}_m \cdot \mathbf{v}_m dV_{\mathbf{X}} + \int_{\partial\Omega} (\det \mathbf{F}) \mathbf{v}_m \cdot \mathbf{S}_m \mathbf{F}^{-T} \mathbf{N} dA_{\mathbf{X}}, \end{aligned}$$

which, by definition of the material velocity field $\dot{\boldsymbol{\varphi}}$ and the first Piola-Kirchhoff stress field $\mathbf{P}$, yields

$$\begin{aligned} \mathcal{P}[\Omega_t] &= \int_{\Omega} \rho_0 \mathbf{b}_m \cdot \dot{\boldsymbol{\varphi}} dV_{\mathbf{X}} + \int_{\partial\Omega} \dot{\boldsymbol{\varphi}} \cdot \mathbf{P} \mathbf{N} dA_{\mathbf{X}} \\ &= \int_{\Omega} \rho_0 \mathbf{b}_m \cdot \dot{\boldsymbol{\varphi}} dV_{\mathbf{X}} + \int_{\partial\Omega} \mathbf{P}^T \dot{\boldsymbol{\varphi}} \cdot \mathbf{N} dA_{\mathbf{X}}. \end{aligned}$$

Furthermore, by use of the Divergence Theorem together with Result 2.11, we obtain

$$\begin{aligned} \mathcal{P}[\Omega_t] &= \int_{\Omega} [\rho_0 \mathbf{b}_m \cdot \dot{\boldsymbol{\varphi}} + \nabla^x \cdot (\mathbf{P}^T \dot{\boldsymbol{\varphi}})] dV_{\mathbf{X}} \\ &= \int_{\Omega} [(\nabla^x \cdot \mathbf{P} + \rho_0 \mathbf{b}_m) \cdot \dot{\boldsymbol{\varphi}} + \mathbf{P} : \nabla^x \dot{\boldsymbol{\varphi}}] dV_{\mathbf{X}}. \quad (5.31) \end{aligned}$$

Taking the time derivative of (5.30) and using Result 5.15 to substitute for $\rho_0 \ddot{\boldsymbol{\varphi}}$ gives

$$\frac{d}{dt} K[\Omega_t] = \int_{\Omega} \rho_0 \dot{\boldsymbol{\varphi}} \cdot \ddot{\boldsymbol{\varphi}} dV_{\mathbf{X}} = \int_{\Omega} (\nabla^x \cdot \mathbf{P} + \rho_0 \mathbf{b}_m) \cdot \dot{\boldsymbol{\varphi}} dV_{\mathbf{X}}.$$

Substituting (5.31) into the right-hand side of this expression, and using the fact that

$$\nabla^x \dot{\boldsymbol{\varphi}}(\mathbf{X}, t) = \nabla^x \frac{\partial}{\partial t} \boldsymbol{\varphi}(\mathbf{X}, t) = \frac{\partial}{\partial t} \nabla^x \boldsymbol{\varphi}(\mathbf{X}, t) = \dot{\mathbf{F}}(\mathbf{X}, t),$$

we arrive at the following result.

Result 5.17 Net Working in Lagrangian Form. Let $\varphi : B \times [0, \infty) \rightarrow \mathbf{E}^3$ be a motion of a continuum body with deformation gradient $\mathbf{F}(\mathbf{X}, t)$ and first Piola–Kirchhoff stress $\mathbf{P}(\mathbf{X}, t)$, and let Ω_t be an arbitrary open subset of B_t with corresponding subset Ω in B . Then

$$\frac{d}{dt}K[\Omega_t] + \int_{\Omega} \mathbf{P} : \dot{\mathbf{F}} \, dV_{\mathbf{X}} = \mathcal{P}[\Omega_t], \quad \forall t \geq 0,$$

where $K[\Omega_t]$ is the total kinetic energy and $\mathcal{P}[\Omega_t]$ is the power of external forces on Ω_t . Thus, in view of (5.14), the net working $\mathcal{W}[\Omega_t]$ is given by

$$\mathcal{W}[\Omega_t] = \int_{\Omega} \mathbf{P} : \dot{\mathbf{F}} \, dV_{\mathbf{X}}, \quad \forall t \geq 0.$$

□

A comparison of Results 5.17 and 5.9 shows that $\mathbf{P} : \dot{\mathbf{F}}$ and $\mathbf{S} : \mathbf{L}$ each provide a measure of the rate of work (power) done by internal forces (stresses) in a continuum body. While the quantity $\mathbf{S} : \mathbf{L}$ measures the stress power per unit volume of the current configuration B_t , the quantity $\mathbf{P} : \dot{\mathbf{F}}$ measures the stress power per unit volume of the reference configuration B (see Exercise 10).

5.4.5 First Law of Thermodynamics

As stated in Axiom 5.3, the balance law of energy for an arbitrary open subset Ω_t of B_t is

$$\frac{d}{dt}U[\Omega_t] = Q[\Omega_t] + \mathcal{W}[\Omega_t], \quad (5.32)$$

where $U[\Omega_t]$ is the internal energy of Ω_t defined as

$$U[\Omega_t] = \int_{\Omega_t} \rho(\mathbf{x}, t) \phi(\mathbf{x}, t) \, dV_{\mathbf{x}},$$

$Q[\Omega_t]$ is the net heating of Ω_t defined as

$$Q[\Omega_t] = \int_{\Omega_t} \rho(\mathbf{x}, t) r(\mathbf{x}, t) \, dV_{\mathbf{x}} - \int_{\partial\Omega_t} \mathbf{q}(\mathbf{x}, t) \cdot \mathbf{n}(\mathbf{x}) \, dA_{\mathbf{x}},$$

and $\mathcal{W}[\Omega_t]$ is the net working of external forces on Ω_t .

Performing a change of variable in the integral for $U[\Omega_t]$ we obtain

$$U[\Omega_t] = \int_{\Omega} \rho_0(\mathbf{X}) \Phi(\mathbf{X}, t) \, dV_{\mathbf{X}}, \quad (5.33)$$

where $\Phi(\mathbf{X}, t) = \phi_m(\mathbf{X}, t)$ is the material description of the spatial internal energy field $\phi(\mathbf{x}, t)$ per unit mass. Similarly, performing a change

of variable in the integrals for $Q[\Omega_t]$ we obtain (omitting the arguments $\mathbf{X}$ and t for brevity)

$$Q[\Omega_t] = \int_{\Omega} \rho_0 r_m \, dV_{\mathbf{X}} - \int_{\partial\Omega} (\det \mathbf{F}) \mathbf{q}_m \cdot \mathbf{F}^{-T} \mathbf{N} \, dA_{\mathbf{X}}.$$

If we introduce a material heat flux vector field by

$$\mathbf{Q}(\mathbf{X}, t) = (\det \mathbf{F}(\mathbf{X}, t)) \mathbf{F}(\mathbf{X}, t)^{-1} \mathbf{q}_m(\mathbf{X}, t), \quad (5.34)$$

and a material heat supply field by

$$R(\mathbf{X}, t) = r_m(\mathbf{X}, t),$$

then the net heating can be written in the convenient form

$$Q[\Omega_t] = \int_{\Omega} \rho_0 R \, dV_{\mathbf{X}} - \int_{\partial\Omega} \mathbf{Q} \cdot \mathbf{N} \, dA_{\mathbf{X}}. \quad (5.35)$$

Substituting (5.35) and (5.33) into (5.32), and using Result 5.17 together with the Divergence Theorem, we obtain the following result.

Result 5.18 Law of Energy in Lagrangian Form. *Let $\varphi : B \times [0, \infty) \rightarrow \mathbf{E}^3$ be a motion of a continuum body with associated material internal energy field $\Phi(\mathbf{X}, t)$, material heat flux vector field $\mathbf{Q}(\mathbf{X}, t)$ and material heat supply field $R(\mathbf{X}, t)$. Then*

$$\rho_0 \dot{\Phi} = \mathbf{P} : \dot{\mathbf{F}} - \nabla^{\mathbf{X}} \cdot \mathbf{Q} + \rho_0 R, \quad \forall \mathbf{X} \in B, \, t \geq 0.$$

□

5.4.6 Second Law of Thermodynamics

As with the balance law for linear momentum, there are various ways to develop the Lagrangian form of the Second Law of Thermodynamics (Clausius–Duhem inequality). We may proceed directly from the integral form in Axiom 5.4. Alternatively, we may integrate the local Eulerian form in Result 5.11 over an arbitrary open subset Ω_t of B_t , change variables to obtain an integral over the corresponding subset Ω of B , and then localize. Here we follow the second approach.

Integration of the expression in Result 5.11 over an arbitrary open subset Ω_t of B_t yields

$$\int_{\Omega_t} \rho \dot{\eta} \, dV_{\mathbf{x}} \geq \int_{\Omega_t} \frac{\rho r}{\theta} \, dV_{\mathbf{x}} - \int_{\partial\Omega_t} \frac{\mathbf{q} \cdot \mathbf{n}}{\theta} \, dA_{\mathbf{x}},$$

where $\mathbf{n}$ is the unit outward normal on $\partial\Omega_t$. By performing a change of variable in the above integrals we obtain

$$\int_{\Omega} \rho_0 \dot{\eta}_m \, dV_{\mathbf{X}} \geq \int_{\Omega} \frac{\rho_0 R}{\Theta} \, dV_{\mathbf{X}} - \int_{\partial\Omega} \frac{\mathbf{Q} \cdot \mathbf{N}}{\Theta} \, dA_{\mathbf{X}},$$

where $\Theta = \theta_m$ is the material description of the spatial temperature field θ , η_m is the material description of the spatial entropy field η , and $\mathbf{Q}$ is the material heat flux vector field defined in (5.34). The Divergence Theorem together with the Localization Theorem then lead to the following result.

Result 5.19 Clausius–Duhem Inequality in Lagrangian Form.

Let $\varphi : B \times [0, \infty) \rightarrow \mathbf{E}^3$ be a motion of a continuum body. Then

$$\rho_0 \dot{\eta}_m \geq \Theta^{-1} \rho_0 R - \nabla^x \cdot (\Theta^{-1} \mathbf{Q}), \quad \forall \mathbf{X} \in B, \, t \geq 0.$$

□

As in the Eulerian case, it is useful to introduce a free energy density per unit mass

$$\Psi(\mathbf{X}, t) = \Phi(\mathbf{X}, t) - \Theta(\mathbf{X}, t) \eta_m(\mathbf{X}, t).$$

Using the free energy, and arguments similar to those leading to Result 5.12, we obtain the following result (see Exercise 11).

Result 5.20 Reduced Clausius–Duhem Inequality in Lagrangian Form. Let $\varphi : B \times [0, \infty) \rightarrow \mathbf{E}^3$ be a motion of a continuum body. Then

$$\rho_0 \dot{\Psi} \leq \mathbf{P} : \dot{\mathbf{F}} - \rho_0 \eta_m \dot{\Theta} - \Theta^{-1} \mathbf{Q} \cdot \nabla^x \Theta, \quad \forall \mathbf{X} \in B, \, t \geq 0.$$

□

All the remarks about the Eulerian formulation of the Clausius–Duhem inequality regarding internal dissipation, flow of heat and free energy carry over to this formulation.

5.4.7 Summary

In the Lagrangian description of the motion of a general continuum body there are 21 basic unknown fields:

$$\varphi_i(\mathbf{X}, t)$$

3 components of motion

$V_i(\mathbf{X}, t)$	3 components of velocity
$P_{ij}(\mathbf{X}, t)$	9 components of stress
$\Theta(\mathbf{X}, t)$	1 material temperature
$Q_i(\mathbf{X}, t)$	3 components of heat flux
$\Phi(\mathbf{X}, t)$	1 internal energy per unit mass
$\eta_m(\mathbf{X}, t)$	1 entropy per unit mass.

To determine these unknown fields we have the following 10 equations:

$V_i = \frac{\partial}{\partial t} \varphi_i$	3 kinematical
$\rho_0 \dot{V}_i = P_{ij,j} + \rho_0 [b_i]_m$	3 linear momentum
$P_{ik} F_{jk} = F_{ik} P_{jk}$	3 independent angular momentum
$\rho_0 \dot{\Phi} = P_{ij} \dot{F}_{ij} - Q_{i,i} + \rho_0 R$	1 energy.

Remarks:

- (1) In contrast with the Eulerian formulation, the mass density field in the Lagrangian formulation is a known quantity. It is just the prescribed density ρ_0 of the body in the reference configuration.
- (2) In many situations the material velocity field $\mathbf{V}$ is not needed explicitly. In this case, the number of unknowns is reduced to 18 ($\mathbf{V}$ disappears), and the number of equations is reduced to 7 (kinematical equations disappear).
- (3) Since the number of unknowns is greater than the number of equations by 11, extra equations are required to close the system. As we will see later, system closure can be achieved by the introduction of constitutive equations which relate $(\boldsymbol{\Sigma}, \mathbf{Q}, \Phi, \eta_m)$ to $(\boldsymbol{\varphi}, \Theta)$, where $\boldsymbol{\Sigma} = \mathbf{F}^{-1} \mathbf{P}$ is the (symmetric) second Piola–Kirchhoff stress field.
- (4) As mentioned earlier, the Second Law (Clausius–Duhem inequality) does not provide an equation with which to determine unknown fields. Instead, we interpret the Second Law as providing a restriction on constitutive relations. In particular, the constitutive relations of a body must guarantee that the Second Law is satisfied in every possible motion of that body (see Exercise 12).
- (5) When thermal effects are neglected, the number of unknowns

is further reduced from 18 to 12 ($\mathbf{Q}$, Φ , η_m and Θ disappear), and the number of equations is reduced from 7 to 6 (balance of energy disappears). In this case, system closure can be obtained by 6 constitutive equations relating Σ to φ . $\square$

5.5 Frame-Indifference

In this section we introduce the idea of a superposed rigid motion and use it to define the concept of frame-indifference. We then state an axiom which asserts that certain fields in continuum mechanics are frame-indifferent, or equivalently, independent of observer. As we will see, this axiom will place strict limitations on the constitutive equations of a material body.

5.5.1 Superposed Rigid Motions

If two observers in different reference frames witness the motion of a body, then the two motions so witnessed, when referred to a common frame, must be related by a superposed rigid motion as defined next.

Definition 5.21 *Two motions $\varphi, \varphi^* : B \times [0, \infty) \rightarrow E^3$ of a body are related by a **superposed rigid motion** if*

$$\varphi^*(\mathbf{X}, t) = \mathbf{Q}(t)\varphi(\mathbf{X}, t) + \mathbf{c}(t), \quad \forall \mathbf{X} \in B, t \geq 0, \quad (5.36)$$

for some rotation $\mathbf{Q}(t)$ and vector $\mathbf{c}(t)$. $\square$

In the above definition, the functions $\mathbf{Q}(t)$ and $\mathbf{c}(t)$ describe the movement of one observer relative to the other. In particular, $\mathbf{Q}(t)$ describes the relative rotation, whereas $\mathbf{c}(t)$ describes the relative translation. Thus different pairs of observers are described by different functions $\mathbf{Q}(t)$ and $\mathbf{c}(t)$. Throughout our developments we implicitly assume that any two observers measure time relative to the same clock. A more general relation between φ and φ^* would be necessary to represent observers with different clocks.

The relation between various measures of strain and rate of strain as seen by two different observers can be derived from (5.36). In particular, let $\mathbf{v}(\mathbf{x}, t)$ and $\mathbf{F}(\mathbf{X}, t)$ be the spatial velocity and deformation gradient fields associated with φ , and let $\mathbf{v}^*(\mathbf{x}^*, t)$ and $\mathbf{F}^*(\mathbf{X}, t)$ be the corresponding quantities associated with φ^* . Moreover, let $\mathbf{C}$, $\mathbf{VR}$ and $\mathbf{RU}$ be the Cauchy–Green strain tensor and the left and right polar

decompositions associated with $\mathbf{F}$, and let $\mathbf{C}^*$, $\mathbf{V}^*\mathbf{R}^*$ and $\mathbf{R}^*\mathbf{U}^*$ be the corresponding quantities associated with $\mathbf{F}^*$. Finally, let $\mathbf{L}$ be the rate of strain field associated with $\mathbf{v}$, and let $\mathbf{L}^*$ be the rate of strain associated with $\mathbf{v}^*$. Then from (5.36) we deduce the following result (see Exercise 13).

Result 5.22 *Effect of Superposed Rigid Motion.* The relation between the material fields $(\mathbf{F}, \mathbf{R}, \mathbf{U}, \mathbf{V}, \mathbf{C})$ and $(\mathbf{F}^*, \mathbf{R}^*, \mathbf{U}^*, \mathbf{V}^*, \mathbf{C}^*)$ is given by

$$\mathbf{F}^* = \mathbf{Q}\mathbf{F}, \quad \mathbf{R}^* = \mathbf{Q}\mathbf{R},$$

$$\mathbf{U}^* = \mathbf{U}, \quad \mathbf{V}^* = \mathbf{Q}\mathbf{V}\mathbf{Q}^T, \quad \mathbf{C}^* = \mathbf{C},$$

for all $\mathbf{X} \in B$ and $t \geq 0$. The relation between the spatial fields $(\nabla^x \mathbf{v}, \mathbf{L})$ and $(\nabla^{x^*} \mathbf{v}^*, \mathbf{L}^*)$ is given by

$$\nabla^{x^*} \mathbf{v}^*(\mathbf{x}^*, t) \Big|_{\mathbf{x}^* = \mathbf{g}(\mathbf{x}, t)} = \mathbf{Q}(t) \nabla^x \mathbf{v}(\mathbf{x}, t) \mathbf{Q}(t)^T + \dot{\mathbf{Q}}(t) \mathbf{Q}(t)^T,$$

$$\mathbf{L}^*(\mathbf{x}^*, t) \Big|_{\mathbf{x}^* = \mathbf{g}(\mathbf{x}, t)} = \mathbf{Q}(t) \mathbf{L}(\mathbf{x}, t) \mathbf{Q}(t)^T,$$

for all $\mathbf{x} \in B_t$ and $t \geq 0$. Here $\dot{\mathbf{Q}}(t)$ denotes the derivative of $\mathbf{Q}(t)$. $\square$

5.5.2 Axiom of Frame-Indifference

At any time $t \geq 0$, let B_t denote the current placement of B under the motion φ , let B_t^* denote the current placement under φ^* , and let $\mathbf{g}_t : B_t \rightarrow B_t^*$ denote the rigid motion defined by (5.36), that is

$$\mathbf{g}_t(\mathbf{x}) = \mathbf{g}(\mathbf{x}, t) = \mathbf{Q}(t)\mathbf{x} + \mathbf{c}(t).$$

Let $\phi(\mathbf{x}, t)$ be a scalar field, $\mathbf{w}(\mathbf{x}, t)$ a vector field and $\mathbf{S}(\mathbf{x}, t)$ a second-order tensor field associated with the body in its placement B_t whose points we denote by $\mathbf{x}$. Moreover, let $\phi^*(\mathbf{x}^*, t)$, $\mathbf{w}^*(\mathbf{x}^*, t)$ and $\mathbf{S}^*(\mathbf{x}^*, t)$ be the corresponding fields associated with the body in its placement B_t^* whose points we denote by $\mathbf{x}^*$.

Definition 5.23 The fields ϕ , $\mathbf{w}$ and $\mathbf{S}$ are called **frame-indifferent** if for all superposed rigid motions $\mathbf{g}_t : B_t \rightarrow B_t^*$ we have

$$\phi^*(\mathbf{x}^*, t) = \phi(\mathbf{x}, t),$$

$$\mathbf{w}^*(\mathbf{x}^*, t) = \mathbf{Q}(t)\mathbf{w}(\mathbf{x}, t),$$

$$\mathbf{S}^*(\mathbf{x}^*, t) = \mathbf{Q}(t)\mathbf{S}(\mathbf{x}, t)\mathbf{Q}(t)^T,$$

for all $\mathbf{x} \in B_t$ and $t \geq 0$. Here $\mathbf{x}^* = \mathbf{g}(\mathbf{x}, t)$, that is

$$\mathbf{x}^* = \mathbf{Q}(t)\mathbf{x} + \mathbf{c}(t).$$

□

The above notion of frame-indifference captures the idea that some physical quantities associated with a body are inherent to the body and independent of observer. For example, any two observers in different reference frames should agree on the mass density and temperature of a continuum body. They should also agree, after an appropriate change of basis, on the components of the heat flux vector and stress tensor fields. These ideas are made precise by saying that the mass density, temperature, heat flux and Cauchy stress fields are all frame-indifferent in the sense of Definition 5.23. Not all fields, however, are frame-indifferent. For example, two observers in relative motion will generally disagree on the velocity and acceleration fields in a body. Common experience and intuition suggest the following assumption.

Axiom 5.24 Material Frame-Indifference. *The spatial mass density $\rho(\mathbf{x}, t)$, temperature $\theta(\mathbf{x}, t)$, entropy per unit mass $\eta(\mathbf{x}, t)$, internal energy per unit mass $\phi(\mathbf{x}, t)$, Cauchy stress $\mathbf{S}(\mathbf{x}, t)$ and Fourier heat flux $\mathbf{q}(\mathbf{x}, t)$ are all frame-indifferent fields.* □

The above axiom places strict limitations on the constitutive equations used to describe the material properties of a given body. For example, suppose the internal energy, heat flux and Cauchy stress are modeled by constitutive equations of the form

$$\begin{aligned}\phi(\mathbf{x}, t) &= \widehat{\phi}(\rho(\mathbf{x}, t), \theta(\mathbf{x}, t), \mathbf{L}(\mathbf{x}, t)), \\ \mathbf{q}(\mathbf{x}, t) &= \widehat{\mathbf{q}}(\rho(\mathbf{x}, t), \theta(\mathbf{x}, t), \mathbf{L}(\mathbf{x}, t)), \\ \mathbf{S}(\mathbf{x}, t) &= \widehat{\mathbf{S}}(\rho(\mathbf{x}, t), \theta(\mathbf{x}, t), \mathbf{L}(\mathbf{x}, t)),\end{aligned}$$

where $\widehat{\phi}$, $\widehat{\mathbf{q}}$ and $\widehat{\mathbf{S}}$ are given functions. Then in order to comply with Axiom 5.24 these functions must have the property (omitting the arguments $\mathbf{x}$, $\mathbf{x}^*$ and t for brevity)

$$\begin{aligned}\widehat{\phi}(\rho^*, \theta^*, \mathbf{L}^*) &= \widehat{\phi}(\rho, \theta, \mathbf{L}), \\ \widehat{\mathbf{q}}(\rho^*, \theta^*, \mathbf{L}^*) &= \mathbf{Q}\widehat{\mathbf{q}}(\rho, \theta, \mathbf{L}), \\ \widehat{\mathbf{S}}(\rho^*, \theta^*, \mathbf{L}^*) &= \mathbf{Q}\widehat{\mathbf{S}}(\rho, \theta, \mathbf{L})\mathbf{Q}^T.\end{aligned}$$

Since $\rho^* = \rho$ and $\theta^* = \theta$ by Axiom 5.24, and $\mathbf{L}^* = \mathbf{Q}\mathbf{L}\mathbf{Q}^T$ by Result 5.22, we deduce that the functions $\hat{\phi}$, $\hat{q}$ and $\hat{\mathbf{S}}$ must satisfy

$$\begin{aligned}\hat{\phi}(\rho, \theta, \mathbf{Q}\mathbf{L}\mathbf{Q}^T) &= \hat{\phi}(\rho, \theta, \mathbf{L}), \\ \hat{q}(\rho, \theta, \mathbf{Q}\mathbf{L}\mathbf{Q}^T) &= \mathbf{Q}\hat{q}(\rho, \theta, \mathbf{L}), \\ \hat{\mathbf{S}}(\rho, \theta, \mathbf{Q}\mathbf{L}\mathbf{Q}^T) &= \mathbf{Q}\hat{\mathbf{S}}(\rho, \theta, \mathbf{L})\mathbf{Q}^T,\end{aligned}$$

for all rotations $\mathbf{Q}$ and all admissible values of ρ , θ and $\mathbf{L}$. Thus the axiom of material frame-indifference imposes severe restrictions on the possible forms of $\hat{\phi}$, $\hat{q}$ and $\hat{\mathbf{S}}$. Functions violating these restrictions would deliver different results when employed by different observers (see Exercise 14). For this reason, frame-indifference is imposed on constitutive equations either exactly or, for linearized models, approximately.

5.6 Material Constraints

Physical experience tells us that certain materials greatly resist certain types of deformations. For example, liquids such as water greatly resist changes in volume, but are otherwise very deformable. We usually model such materials by placing a-priori restrictions or constraints on their motions.

Definition 5.25 *A continuum body is said to be subject to a simple material or internal constraint if every motion $\varphi : B \times [0, \infty) \rightarrow \mathbf{E}^3$ must satisfy an equation of the form*

$$\gamma(\mathbf{F}(\mathbf{X}, t)) = 0, \quad \forall \mathbf{X} \in B, \quad t \geq 0, \quad (5.37)$$

where $\mathbf{F}(\mathbf{X}, t)$ is the deformation gradient and $\gamma : \mathcal{V}^2 \rightarrow \mathbf{R}$ is a specified function. □

As alluded to above, resistance to volume change is an important example of a material constraint. Materials which admit no volume change are called **incompressible**. Such materials can experience only volume-preserving motions. In view of Result 4.12, an appropriate constraint function for an incompressible material is

$$\gamma(\mathbf{F}) = \det \mathbf{F} - 1. \quad (5.38)$$

When this function is substituted into (5.37) we obtain $\det \mathbf{F}(\mathbf{X}, t) = 1$ for all $\mathbf{X} \in B$ and $t \geq 0$, which implies that the local volume of material in a neighborhood of each point is constant in time. Result 4.12 also shows that an equivalent formulation of this constraint is $\nabla^x \cdot \mathbf{v}(\mathbf{x}, t) = 0$

for all $\mathbf{x} \in B_t$ and $t \geq 0$. In particular, this divergence condition implies $\det \mathbf{F}(\mathbf{X}, t) = 1$ for all $t > 0$ provided it holds at $t = 0$.

Material constraints must be enforced or maintained by appropriate stresses. While many different systems of stresses could presumably enforce a given constraint, we make the following assumption.

Axiom 5.26 Stress Fields in Constrained Materials. *The first Piola–Kirchhoff stress field $\mathbf{P}(\mathbf{X}, t)$ in a continuum body subject to a material constraint $\gamma(\mathbf{F}(\mathbf{X}, t)) = 0$ can be decomposed into two parts*

$$\mathbf{P}(\mathbf{X}, t) = \mathbf{P}^{(r)}(\mathbf{X}, t) + \mathbf{P}^{(a)}(\mathbf{X}, t).$$

Here $\mathbf{P}^{(a)}(\mathbf{X}, t)$ is an active stress field determined by a constitutive equation, and $\mathbf{P}^{(r)}(\mathbf{X}, t)$ is a reactive stress field whose stress power is zero in all possible motions, that is

$$\mathbf{P}^{(r)}(\mathbf{X}, t) : \dot{\mathbf{F}}(\mathbf{X}, t) = 0, \quad \forall \mathbf{X} \in B, t \geq 0. \quad (5.39)$$

□

From (5.39) we can deduce the most general form of the reactive stress $\mathbf{P}^{(r)}$ associated with a given constraint. The condition that $\gamma(\mathbf{F}(\mathbf{X}, t))$ be constant in time for all $\mathbf{X} \in B$ is equivalent to the condition

$$0 = \frac{\partial}{\partial t} \gamma(\mathbf{F}(\mathbf{X}, t)) = D\gamma(\mathbf{F}(\mathbf{X}, t)) : \dot{\mathbf{F}}(\mathbf{X}, t), \quad \forall \mathbf{X} \in B, t \geq 0.$$

Thus all motions satisfying the constraint have the property that $\dot{\mathbf{F}}$ is orthogonal to $D\gamma(\mathbf{F})$ with respect to the standard inner product in the space of second-order tensors. In particular, $\dot{\mathbf{F}}$ must lie in the hyperplane orthogonal to $D\gamma(\mathbf{F})$ for each $\mathbf{X} \in B$ and $t \geq 0$. Here $D\gamma(\mathbf{F})$ denotes the derivative of γ at $\mathbf{F}$ (see Definition 2.20).

The fact that (5.39) must hold for all possible motions satisfying the constraint implies that $\mathbf{P}^{(r)}$ must be orthogonal to the hyperplane defined by $D\gamma(\mathbf{F})$. In particular, $\mathbf{P}^{(r)}$ must be parallel to $D\gamma(\mathbf{F})$ for each $\mathbf{X} \in B$ and $t \geq 0$. Thus the most general form for the reactive stress is

$$\mathbf{P}^{(r)}(\mathbf{X}, t) = q(\mathbf{X}, t) D\gamma(\mathbf{F}(\mathbf{X}, t)),$$

where $q(\mathbf{X}, t)$ is an unknown scalar field referred to as the **multiplier**. It is the unknown part of the reactive stress that enforces the constraint $\gamma(\mathbf{F}(\mathbf{X}, t)) = 0$. For example, for incompressible materials defined by (5.38), we deduce (see Result 2.22)

$$\mathbf{P}^{(r)}(\mathbf{X}, t) = q(\mathbf{X}, t) \det(\mathbf{F}(\mathbf{X}, t)) \mathbf{F}(\mathbf{X}, t)^{-T}. \quad (5.40)$$

Using Definition 5.14 we can translate the content of Axiom 5.26 from the first Piola–Kirchhoff to the Cauchy stress field. In particular, the Cauchy stress in a body subject to a material constraint can be decomposed as

$$\mathbf{S}(\mathbf{x}, t) = \mathbf{S}^{(r)}(\mathbf{x}, t) + \mathbf{S}^{(a)}(\mathbf{x}, t),$$

where $\mathbf{S}^{(r)}$ is a reactive Cauchy stress corresponding to $\mathbf{P}^{(r)}$, and $\mathbf{S}^{(a)}$ is an active Cauchy stress corresponding to $\mathbf{P}^{(a)}$. For example, for incompressible materials, we deduce from (5.40) that the reactive Cauchy stress takes the form

$$\mathbf{S}^{(r)}(\mathbf{x}, t) = -p(\mathbf{x}, t)\mathbf{I},$$

where $-p(\mathbf{x}, t) = q_s(\mathbf{x}, t)$ is the spatial description of the material field $q(\mathbf{X}, t)$. Thus the reactive Cauchy stress is spherical (see Chapter 3) and the multiplier field can be interpreted as a pressure.

Axiom 5.24 asserts that the Cauchy stress field in a body is frame-indifferent. In the earlier case of a body with no constraint, we showed that this axiom places a restriction on the constitutive equation for $\mathbf{S}(\mathbf{x}, t)$. In the present case of a body with a constraint, we interpret this axiom as placing a restriction on the possible forms of the constraint function γ and the constitutive equation for the active stress $\mathbf{S}^{(a)}$. In particular, we will always assume that $\gamma(\mathbf{F}(\mathbf{X}, t))$ and $\mathbf{S}^{(a)}(\mathbf{x}, t)$ are both frame-indifferent. These assumptions, together with the frame-indifference of $\mathbf{S}(\mathbf{x}, t)$, imply that the multiplier field is frame-indifferent.

5.7 Isothermal Considerations

Many classic models of fluids and solids are purely mechanical. Here we introduce the notion of an isothermal process as a means of describing such models within the framework of this chapter, and discuss the balance laws relevant to the isothermal modeling of continuum bodies.

Definition 5.27 *By a thermo-mechanical process for a continuum body we mean a pair of functions (φ, Θ) , where $\varphi(\mathbf{X}, t)$ is an admissible motion and $\Theta(\mathbf{X}, t)$ is a material temperature field. A process is called **isothermal** if $\Theta(\mathbf{X}, t)$ is constant independent of $\mathbf{X}$ and t , equivalently, if the spatial temperature $\theta(\mathbf{x}, t)$ is constant independent of $\mathbf{x}$ and t . $\square$*

Our basic assumption in the isothermal modeling of a continuum body is that the body can experience only isothermal processes. In particular, for every motion φ we assume the material temperature Θ is constant for

all $\mathbf{X} \in B$ and $t \geq 0$. This assumption is physically reasonable for bodies with negligible internal and external sources of heat. Internal sources of heat are expected to be negligible when the body is subject to mild rates of strain. External sources of heat are expected to be negligible when the temperature of the body environment is nearly uniform, constant and equal to that of the body. When these conditions are violated the isothermal assumption is no longer expected to be valid.

The balance laws of mass, linear momentum and angular momentum and the axiom of material frame-indifference are fundamental in the isothermal modeling of continuum bodies. The balance law of energy is not directly relevant because it contains quantities, namely the internal energy, heat flux vector and heat supply, which are typically left unspecified in isothermal models. If any two of these quantities were to be known, then the energy balance law could be used to gain information about the third. For simplicity, in our analysis of isothermal models in Chapters 6 and 7 we leave all thermal quantities including entropy unspecified and make no reference to the energy balance law. At a first glance it appears that the Clausius–Duhem inequality (Second Law of Thermodynamics) is also not directly relevant. However, we show below that it leads to a purely mechanical energy inequality for an important class of bodies.

Definition 5.28 *A thermo-mechanical process (φ, Θ) for a body is said to be **closed** in an interval $[t_0, t_1]$ if $\varphi(\mathbf{X}, t_1) = \varphi(\mathbf{X}, t_0)$, $\dot{\varphi}(\mathbf{X}, t_1) = \dot{\varphi}(\mathbf{X}, t_0)$ and $\Theta(\mathbf{X}, t_1) = \Theta(\mathbf{X}, t_0)$ for all $\mathbf{X} \in B$. A body is called **energetically passive** if for every closed process its material free energy density satisfies $\Psi(\mathbf{X}, t_1) - \Psi(\mathbf{X}, t_0) \geq 0$ for all $\mathbf{X} \in B$. $\square$*

Thus a process is closed in an interval $[t_0, t_1]$ if all particles in a body have the same position, velocity and temperature at time t_1 as they had at time t_0 . Moreover, a body is energetically passive if the net change in its free energy density is non-negative in any closed process. For example, when Ψ depends only on the current value of the fields φ , $\dot{\varphi}$ and Θ , and not on their past history, the net change in Ψ is zero. The next result is a consequence of the reduced Clausius–Duhem inequality in either Lagrangian or Eulerian form (Results 5.20 and 5.12).

Result 5.29 Mechanical Energy Inequality. *Consider an energetically passive body with reference configuration B . Then for any closed*

isothermal process in an interval $[t_0, t_1]$ we must have

$$\int_{t_0}^{t_1} \mathbf{P}(\mathbf{X}, t) : \dot{\mathbf{F}}(\mathbf{X}, t) dt \geq 0, \quad \forall \mathbf{X} \in B, \quad (5.41)$$

where $\mathbf{P}(\mathbf{X}, t)$ is the first Piola–Kirchhoff stress field and $\mathbf{F}(\mathbf{X}, t)$ is the deformation gradient. Equivalently

$$\int_{t_0}^{t_1} (\det \mathbf{F}(\mathbf{X}, t)) \mathbf{S}_m(\mathbf{X}, t) : \mathbf{L}_m(\mathbf{X}, t) dt \geq 0, \quad \forall \mathbf{X} \in B, \quad (5.42)$$

where $\mathbf{S}(\mathbf{x}, t)$ is the Cauchy stress field and $\mathbf{L}(\mathbf{x}, t)$ is the rate of strain field. $\square$

Proof For any thermo-mechanical process the reduced Clausius–Duhem inequality in Lagrangian form (Result 5.20) states

$$\rho_0 \dot{\Psi} \leq \mathbf{P} : \dot{\mathbf{F}} - \rho_0 \eta_m \dot{\Theta} - \Theta^{-1} \mathbf{Q} \cdot \nabla^x \Theta, \quad \forall \mathbf{X} \in B, t \geq 0.$$

If the process is isothermal, we get $\dot{\Theta} = 0$ and $\nabla^x \Theta = \mathbf{0}$, and integrating over a time interval $[t_0, t_1]$ gives

$$\rho_0(\mathbf{X})[\Psi(\mathbf{X}, t_1) - \Psi(\mathbf{X}, t_0)] \leq \int_{t_0}^{t_1} \mathbf{P}(\mathbf{X}, t) : \dot{\mathbf{F}}(\mathbf{X}, t) dt.$$

Additionally, if the process is closed and the body is energetically passive, the left-hand side is non-negative, which yields the result in (5.41). Similarly, for any process the reduced Clausius–Duhem inequality in Eulerian form (Result 5.12) states

$$\rho \dot{\psi} \leq \mathbf{S} : \mathbf{L} - \rho \eta \dot{\theta} - \theta^{-1} \mathbf{q} \cdot \nabla^x \theta, \quad \forall \mathbf{x} \in B_t, t \geq 0.$$

If the process is isothermal, we get $\dot{\theta} = 0$ and $\nabla^x \theta = \mathbf{0}$, which implies

$$\rho \dot{\psi} \leq \mathbf{S} : \mathbf{L}, \quad \forall \mathbf{x} \in B_t, t \geq 0.$$

In material form this reads

$$\rho_m \dot{\Psi} \leq \mathbf{S}_m : \mathbf{L}_m, \quad \forall \mathbf{X} \in B, t \geq 0,$$

where $\Psi = \psi_m$. By conservation of mass (Result 5.13) we have $\rho_m = \rho_0 / \det \mathbf{F}$. Substituting this into the above inequality and integrating as before leads to the result in (5.42). $\square$

Result 5.29 implies that, in any closed isothermal process, an energetically passive body may consume mechanical energy but cannot create

it. In particular, for any open subset Ω_t of B_t we find

$$\int_{t_0}^{t_1} \mathcal{P}[\Omega_t] dt \geq 0, \quad (5.43)$$

where $\mathcal{P}[\Omega_t]$ is the power of external forces on Ω_t . This inequality may be deduced by integrating (5.41) over an arbitrary subset Ω of B and using Result 5.17 together with the definition of a closed process. Equivalently, we may integrate (5.42) and use Result 5.9. The integral in (5.43) corresponds to the work of external forces on Ω_t . The fact that this integral is non-negative implies that Ω_t can only consume energy.

As with the Clausius–Duhem inequality, we view Result 5.29 as a constitutive restriction. In particular, under the assumption that a body is energetically passive, the constitutive equation for the stress field must guarantee that the mechanical energy inequality is satisfied in every closed isothermal process. Thus in the isothermal modeling of continuum bodies we have three basic balance laws pertaining to mass, linear momentum and angular momentum. Moreover, we have two basic constitutive restrictions pertaining to material frame-indifference and the mechanical energy inequality. Notice that the latter applies only to a particular class of bodies and processes.

Bibliographic Notes

Much of the material presented here on the mechanical balance laws can be found in Truesdell (1991) and Gurtin (1981), and much of the material on the thermal balance laws can be found in Truesdell (1984). A comprehensive account of both the mechanical and thermal balance laws can be found in the treatise by Truesdell and Toupin (1960). More concise accounts can be found in Šilhavý (1997), Antman (1995) and Marsden and Hughes (1983) from a mathematical point of view, and in Spencer (1992), Mase (1970) and Malvern (1969) from an engineering point of view.

The definition of entropy and the form of the Second Law presented here is adapted from Truesdell (1984). That text contains a historical overview of these concepts beginning from the classic theory of homogeneous, reversible systems to the modern theory of continuum mechanics. In particular, various issues surrounding the generalization of the Second Law from the classic theory to the present, and its role as a restriction on constitutive relations, are highlighted there. The role of statistical theories in the development of modern thermodynamics is also discussed.

The axiom on material frame-indifference presented here asserts that various quantities associated with a continuum body are invariant under changes of observer as represented by superposed rigid motions. We interpret this axiom as placing restrictions on possible constitutive relations for such quantities. In many texts, this axiom is stated directly as a restriction on the constitutive relations. These two views are mathematically equivalent. The one adopted here is convenient because it does not require a discussion of constitutive theory as a prerequisite. Some authors choose to state the axiom of frame-indifference using superposed motions defined by general orthogonal as opposed to proper orthogonal (rotation) tensors. For a discussion see Truesdell (1991).

Throughout our developments we have assumed that the underlying fields are smooth. A discussion of the balance laws of linear and angular momentum in the non-smooth case can be found in Antman (1995) and references therein. A comprehensive treatment of jump conditions across surfaces of discontinuity in continuum mechanics is given in Šilhavý (1997) and Truesdell and Toupin (1960).

Our presentation on material constraints is adapted from Truesdell (1991). A detailed account of this theory can be found in Antman (1995). The result on the mechanical energy inequality given here is based on the discussion of internal dissipation given in Truesdell (1984).

Exercises

- 5.1 Let $W_{12}(\mathbf{x}, \mathbf{y})$ and $W_{21}(\mathbf{y}, \mathbf{x})$ be scalar-valued functions of two vector variables $\mathbf{x}$ and $\mathbf{y}$, and let $\mathbf{r} = \mathbf{x} - \mathbf{y}$.

(a) Show $\nabla^{\mathbf{x}}|\mathbf{r}| = \frac{\mathbf{r}}{|\mathbf{r}|}$ and $\nabla^{\mathbf{y}}|\mathbf{r}| = -\frac{\mathbf{r}}{|\mathbf{r}|}$.

- (b) Assuming $W_{12}(\mathbf{x}, \mathbf{y}) = \widehat{W}(|\mathbf{r}|) = W_{21}(\mathbf{y}, \mathbf{x})$ show

$$\nabla^{\mathbf{x}}W_{12} = \frac{\widehat{W}'(|\mathbf{r}|)}{|\mathbf{r}|}\mathbf{r}, \quad \nabla^{\mathbf{y}}W_{21} = -\frac{\widehat{W}'(|\mathbf{r}|)}{|\mathbf{r}|}\mathbf{r}.$$

Here $\widehat{W}'(s)$ denotes the derivative of $\widehat{W}(s)$.

- 5.2 Use the results of Exercise 1 to show that (5.1) implies the particle balance laws in (5.3)–(5.6) under the stated assumptions on the interaction energies, namely

$$U_{ij}(\mathbf{x}_i, \mathbf{x}_j) = \widehat{U}_{ij}(|\mathbf{x}_i - \mathbf{x}_j|) = U_{ji}(\mathbf{x}_j, \mathbf{x}_i), \quad (\text{no sum}).$$

- 5.3 Use Axiom 5.2 and the definitions of $\mathbf{j}[\Omega_t]_{\mathbf{z}}$, $\mathbf{l}[\Omega_t]$, $\boldsymbol{\tau}[\Omega_t]_{\mathbf{z}}$ and $\mathbf{r}[\Omega_t]$ to show that the angular momentum of a body Ω_t satisfies

$$\frac{d}{dt}\mathbf{j}[\Omega_t]_{\mathbf{z}} = \boldsymbol{\tau}[\Omega_t]_{\mathbf{z}},$$

where $\mathbf{z}$ is any fixed point.

- 5.4 Use Axioms 5.1 and 5.2 to show that

$$M\dot{\mathbf{x}}_{\text{com}} = \mathbf{l}[\Omega_t], \quad M\ddot{\mathbf{x}}_{\text{com}} = \mathbf{r}[\Omega_t],$$

where M is the mass and $\mathbf{x}_{\text{com}}$ is the center of mass of Ω_t .

Remark: The second result is known as the **Center of Mass Theorem**. It states that the motion of the center of mass of an arbitrary body Ω_t is the same as that of a particle having the same mass as Ω_t , located at the center of mass of Ω_t and subject to the same resultant force as Ω_t .

- 5.5 Use the results of Exercise 4 to extend the result of Exercise 3. In particular, show that the angular momentum of a body Ω_t also satisfies

$$\frac{d}{dt}\mathbf{j}[\Omega_t]_{\mathbf{x}_{\text{com}}} = \boldsymbol{\tau}[\Omega_t]_{\mathbf{x}_{\text{com}}},$$

where $\mathbf{x}_{\text{com}}$ is the center of mass of Ω_t .

- 5.6 Rather than being constant as asserted in Axiom 5.1, suppose the mass of an arbitrary open subset Ω_t of B_t satisfies the growth law

$$\frac{d}{dt}\text{mass}[\Omega_t] = g[\Omega_t], \quad (5.44)$$

where $g[\Omega_t]$ is a net growth rate of the form

$$g[\Omega_t] = \int_{\Omega_t} \gamma(\mathbf{x}, t) \rho(\mathbf{x}, t) dV_{\mathbf{x}}.$$

Here $\gamma(\mathbf{x}, t)$ is a prescribed spatial growth rate per unit mass. Show that the local Eulerian form of (5.44) is

$$\frac{\partial}{\partial t}\rho + \nabla^x \cdot (\rho \mathbf{v}) = \gamma\rho \quad \forall \mathbf{x} \in B_t, t \geq 0.$$

- 5.7 Here we generalize our basic definitions of angular momentum and torque and study the resulting form of the balance of angular momentum equation. In particular, suppose the angular

momentum (about origin) of an arbitrary open subset Ω_t is given by

$$\mathbf{j}[\Omega_t]_o = \int_{\Omega_t} \rho \boldsymbol{\eta} + \mathbf{x} \times \rho \mathbf{v} \, dV_{\mathbf{x}},$$

where $\boldsymbol{\eta}(\mathbf{x}, t)$ is an intrinsic angular momentum field per unit mass, and suppose the resultant torque (about origin) of external influences on Ω_t is given by

$$\boldsymbol{\tau}[\Omega_t]_o = \int_{\Omega_t} \rho \mathbf{h} + \mathbf{x} \times \rho \mathbf{b} \, dV_{\mathbf{x}} + \int_{\partial\Omega_t} \mathbf{m} + \mathbf{x} \times \mathbf{t} \, dA_{\mathbf{x}},$$

where $\mathbf{h}(\mathbf{x}, t)$ is an intrinsic body torque field per unit mass and $\mathbf{m}(\mathbf{x}, t)$ is an intrinsic surface torque field per unit area. Notice that the usual definitions of angular momentum and torque are recovered in the case when $\boldsymbol{\eta}$, $\mathbf{h}$ and $\mathbf{m}$ all vanish. In direct analogy to the relation $\mathbf{t} = \mathbf{S}\mathbf{n}$ for surface traction, we assume $\mathbf{m} = \mathbf{M}\mathbf{n}$ where $\mathbf{M}(\mathbf{x}, t)$ is a second-order tensor field analogous to the Cauchy stress $\mathbf{S}(\mathbf{x}, t)$.

(a) Show that the local Eulerian form of the law of angular momentum in Axiom 5.2 is

$$\rho \dot{\boldsymbol{\eta}} = \nabla^x \cdot \mathbf{M} + \boldsymbol{\xi} + \rho \mathbf{h}, \quad \forall \mathbf{x} \in B_t, \, t \geq 0,$$

where $\boldsymbol{\xi}(\mathbf{x}, t) = \epsilon_{ijk} S_{kj}(\mathbf{x}, t) \mathbf{e}_i$. (Hint: The linear momentum equation is unchanged.)

(b) Assuming $\mathbf{M}$ and $\mathbf{h}$ are identically zero, show that $\dot{\boldsymbol{\eta}}(\mathbf{x}, t) = \mathbf{0}$ if and only if $\mathbf{S}(\mathbf{x}, t)$ is symmetric.

5.8 Let $\boldsymbol{\varphi} : B \times [0, \infty) \rightarrow \mathbf{E}^3$ be a motion of a continuum body with spatial mass density field $\rho(\mathbf{x}, t)$ and spatial velocity field $\mathbf{v}(\mathbf{x}, t)$. Let $\psi(\mathbf{x}, t)$ and $\phi(\mathbf{x}, t)$ be arbitrary scalar fields and let $\mathbf{w}(\mathbf{x}, t)$ be an arbitrary vector field defined on B_t .

(a) Show $(\ln \psi)^\bullet = \psi^{-1} \dot{\psi}$ and $(\psi \phi)^\bullet = \dot{\psi} \phi + \psi \dot{\phi}$, where a dot denotes the total or material time derivative.

(b) Use part (a) and Result 5.5 to show

$$(\rho^{-1} \phi)^\bullet = \rho^{-1} \left[\frac{\partial}{\partial t} \phi + \nabla^x \cdot (\phi \mathbf{v}) \right].$$

(c) Use the result in (b) to show

$$(\rho^{-1} \mathbf{w})^\bullet = \rho^{-1} \left[\frac{\partial}{\partial t} \mathbf{w} + \nabla^x \cdot (\mathbf{w} \otimes \mathbf{v}) \right].$$

- 5.9 Consider a material model where the Cauchy stress tensor is given by

$$\mathbf{S}(\mathbf{x}, t) = -p(\mathbf{x}, t)\mathbf{I}.$$

Here $p(\mathbf{x}, t)$ is a scalar field called pressure. Show that Results 5.5, 5.7 and 5.10 imply

$$\begin{aligned}\frac{\partial \rho}{\partial t} + \nabla^x \cdot (\rho \mathbf{v}) &= 0, \\ \frac{\partial}{\partial t}(\rho \mathbf{v}) + \nabla^x \cdot (\rho \mathbf{v} \otimes \mathbf{v} + p\mathbf{I}) &= \rho \mathbf{b}, \\ \frac{\partial}{\partial t}(\rho E) + \nabla^x \cdot (\rho E \mathbf{v} + p\mathbf{v}) &= \rho r - \nabla^x \cdot \mathbf{q} + \rho \mathbf{b} \cdot \mathbf{v},\end{aligned}$$

where $E = \phi + \frac{1}{2}|\mathbf{v}|^2$ is the total energy density field. Notice that, once constitutive equations relating $(p, \mathbf{q}, \phi)$ to $(\rho, \mathbf{v}, \theta)$ are specified, we obtain a closed system of equations for $(\rho, \mathbf{v}, \theta)$.

- 5.10 Show that the stress power $\mathbf{P} : \dot{\mathbf{F}}$ per unit reference volume can be written as

$$\mathbf{P} : \dot{\mathbf{F}} = \frac{1}{2} \boldsymbol{\Sigma} : \dot{\mathbf{C}},$$

where $\boldsymbol{\Sigma}$ is the second Piola–Kirchhoff stress tensor and $\mathbf{C}$ is the right Cauchy strain tensor (see Chapter 4).

- 5.11 Derive Result 5.20.

- 5.12 Here we study the energy balance equation in Result 5.18 and the Clausius–Duhem inequality in Result 5.20 for a body undergoing an arbitrary rigid motion

$$\boldsymbol{\varphi}(\mathbf{X}, t) = \mathbf{A}(t)\mathbf{X} + \mathbf{c}(t), \quad \forall \mathbf{X} \in B, \quad t \geq 0,$$

where $\mathbf{A}(t)$ is a rotation and $\mathbf{c}(t)$ is a vector. Notice that $\mathbf{A}(t) = \mathbf{I}$ and $\mathbf{c}(t) = \mathbf{0}$ for all $t \geq 0$ corresponds to the case of a body at rest in its reference configuration B .

(a) Use the result of Exercise 10 to show that the stress power $\mathbf{P} : \dot{\mathbf{F}}$ per unit reference volume vanishes identically for an arbitrary rigid motion.

(b) Suppose the internal energy density and heat flux fields are given by constitutive relations of the form $\Phi = \alpha\Theta$ and $\mathbf{Q} =$

$-\kappa \nabla^x \Theta$, where α and κ are scalar constants. Show that the energy balance equation reduces to

$$\rho_0 \alpha \frac{\partial \Theta}{\partial t} = \kappa \Delta^x \Theta + \rho_0 R, \quad \forall \mathbf{X} \in B, t \geq 0.$$

This is a linear partial differential equation for $\Theta(\mathbf{X}, t)$ known as the **Heat Equation**.

(c) Suppose the constitutive relation for Φ is dropped and replaced by constitutive relations for the free energy Ψ and entropy η_m of the form

$$\Psi = \hat{\Psi}(\Theta), \quad \eta_m = -\frac{d\hat{\Psi}}{d\Theta}(\Theta),$$

where $\hat{\Psi}$ is a given function. In this case show that the energy balance equation reduces to

$$-\rho_0 \Theta \frac{d^2 \hat{\Psi}}{d\Theta^2}(\Theta) \frac{\partial \Theta}{\partial t} = \kappa \Delta^x \Theta + \rho_0 R, \quad \forall \mathbf{X} \in B, t \geq 0.$$

This is a nonlinear partial differential equation for $\Theta(\mathbf{X}, t)$.

(d) Assuming constitutive relations as in part (c), show that the Clausius–Duhem inequality in Result 5.20 is satisfied for arbitrary rigid motions and temperature fields only if $\kappa \geq 0$. Thus a model with $\kappa < 0$ would violate the Second Law of Thermodynamics.

5.13 Derive Result 5.22.

5.14 With the aid of Result 5.22 determine which of the following constitutive equations for the Cauchy stress field $\mathbf{S}$ are frame-indifferent. Here μ , λ and γ are arbitrary constants.

$$(a) \mathbf{S} = 2\mu \nabla^x \mathbf{v}. \quad (b) \mathbf{S} = \mu \mathbf{L}. \quad (c) \mathbf{S} = \lambda \rho^\gamma \mathbf{I}.$$

5.15 Consider a material body with spatial velocity field $\mathbf{v}$ and associated spin tensor field $\mathbf{W} = \frac{1}{2}(\nabla^x \mathbf{v} - \nabla^x \mathbf{v}^T)$. Let $\mathbf{T}$ be an arbitrary frame-indifferent spatial tensor field on the body, so $\mathbf{T}^* = \mathbf{Q} \mathbf{T} \mathbf{Q}^T$ for any superposed rigid motion defined by $\mathbf{Q}(t)$. Let $\mathbf{J}_{\text{ma}}(\mathbf{T})$ denote the material or total time derivative of $\mathbf{T}$ defined by

$$\mathbf{J}_{\text{ma}}(\mathbf{T}) = \dot{\mathbf{T}},$$

and let $\mathbf{J}_{\text{co}}(\mathbf{T}, \mathbf{W})$ denote the co-rotational or Jaumann time

derivative of $\mathbf{T}$ with respect to $\mathbf{W}$ defined by (see Chapter 2, Exercise 14)

$$\mathbf{J}_{\text{co}}(\mathbf{T}, \mathbf{W}) = \dot{\mathbf{T}} + \mathbf{W}^T \mathbf{T} + \mathbf{T} \mathbf{W}.$$

(a) For any superposed rigid motion defined by a rotation tensor $\mathbf{Q}(t)$ show that

$$\mathbf{W}^* = \mathbf{Q} \mathbf{W} \mathbf{Q}^T + \dot{\mathbf{Q}} \mathbf{Q}^T.$$

(b) Show that $\mathbf{J}_{\text{co}}(\mathbf{T}, \mathbf{W})$ is frame-indifferent in the sense that

$$\mathbf{J}_{\text{co}}(\mathbf{T}^*, \mathbf{W}^*) = \mathbf{Q} \mathbf{J}_{\text{co}}(\mathbf{T}, \mathbf{W}) \mathbf{Q}^T,$$

whereas $\mathbf{J}_{\text{ma}}(\mathbf{T})$ is not frame-indifferent in the sense that

$$\mathbf{J}_{\text{ma}}(\mathbf{T}^*) \neq \mathbf{Q} \mathbf{J}_{\text{ma}}(\mathbf{T}) \mathbf{Q}^T.$$

Answers to Selected Exercises

5.1 (a) In any coordinate frame $\{\mathbf{e}_k\}$ we have $|\mathbf{r}|^2 = \sum_{q=1}^3 (x_q - y_q)^2$, which gives

$$\frac{\partial |\mathbf{r}|^2}{\partial x_k} = 2(x_k - y_k).$$

Moreover, by the chain rule, the left-hand side is $2|\mathbf{r}|\partial|\mathbf{r}|/\partial x_k$. Thus

$$\frac{\partial |\mathbf{r}|}{\partial x_k} = \frac{(x_k - y_k)}{|\mathbf{r}|} \quad \text{and} \quad \nabla^{\mathbf{x}} |\mathbf{r}| = \sum_{k=1}^3 \frac{\partial |\mathbf{r}|}{\partial x_k} \mathbf{e}_k = \frac{\mathbf{r}}{|\mathbf{r}|}.$$

The second result follows in a similar way, namely

$$\frac{\partial |\mathbf{r}|}{\partial y_k} = -\frac{(x_k - y_k)}{|\mathbf{r}|} \quad \text{and} \quad \nabla^{\mathbf{y}} |\mathbf{r}| = \sum_{k=1}^3 \frac{\partial |\mathbf{r}|}{\partial y_k} \mathbf{e}_k = -\frac{\mathbf{r}}{|\mathbf{r}|}.$$

(b) From the relation $W_{12}(\mathbf{x}, \mathbf{y}) = \widehat{W}(|\mathbf{r}|)$ and the chain rule we get

$$\nabla^{\mathbf{x}} W_{12} = \sum_{k=1}^3 \frac{\partial W_{12}}{\partial x_k} \mathbf{e}_k = \sum_{k=1}^3 \widehat{W}' \frac{\partial |\mathbf{r}|}{\partial x_k} \mathbf{e}_k = \frac{\widehat{W}'}{|\mathbf{r}|} \mathbf{r}.$$

Similarly, since $W_{21}(\mathbf{y}, \mathbf{x}) = \widehat{W}(|\mathbf{r}|)$ we get

$$\nabla^{\mathbf{y}} W_{21} = \sum_{k=1}^3 \frac{\partial W_{21}}{\partial y_k} \mathbf{e}_k = \sum_{k=1}^3 \widehat{W}' \frac{\partial |\mathbf{r}|}{\partial y_k} \mathbf{e}_k = -\frac{\widehat{W}'}{|\mathbf{r}|} \mathbf{r}.$$